

RISK ADJUSTMENT AND TRADING STRATEGIES

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Abstract

In this paper, we assess the profitability of momentum trading strategies using a stochastic discount factor approach. We estimate the stochastic discount factor from a set of basis assets, assuming that the law of one price (or, alternatively, no-arbitrage) holds, and simultaneously assess the risk-adjusted performance of the trading strategies. Our method has the advantages that we need not specify (and therefore perhaps misspecify) a particular parametric model to assess performance, and the method can easily encompass conditional or time-varying moments. In unconditional tests, some profitability of the trading strategies remains relative to the basis asset benchmark, although the level of momentum profits declines substantially after risk adjustment. When expectations are allowed to vary conditional on a limited set of public information, no abnormal profits remain. This result contrasts sharply with those observed when a CAPM benchmark is used. To understand this difference, we investigate the relation between the risk of momentum strategies and the magnitude of the risk premium. We show analytically that the risk of a momentum strategy should be increasing with respect to the market risk premium. Empirically, while the risk measures of the strategies, estimated relative to the stochastic discount factor, behave as predicted, market betas do not. This suggests that the use of CAPM benchmarks to assess the risk-adjusted performance of momentum strategies may lead to incorrect inferences.

1 Introduction

A large number of empirical papers document the ability of investors to achieve abnormal returns through the use of simple trading strategies based on historical returns. Recently, Jegadeesh and Titman (1993) show that, over intermediate horizons of three to twelve months, a portfolio that purchases past winners and sells past losers has a positive abnormal return. The evidence that such a ‘momentum’ strategy exhibits abnormal performance has received a great deal of attention. Some authors have proposed risk-based explanations of these apparent profits [see, e.g., Fama and French (1996)], but to date the evidence that the returns of these strategies is related to identifiable risk measures is mixed. In response, several papers have proposed new theories to explain, among other stylized facts, momentum strategy profits. These papers rely on psychological factors; for example, Barberis, Shleifer and Vishny (1998) build a model which assumes factors such as representativeness and conservatism, while Daniel, Hirshleifer and Subramanyam (1998) rely on overconfidence and self-attribution.

In this paper, we re-assess the risk-based explanations for the profitability of momentum trading strategies by measuring these profits against a nonparametric benchmark. The motivation for this analysis is straightforward. First, existing papers which examine the risk-adjusted performance of trading strategies rely, by necessity, on a particular pricing model or models to measure risk and therefore abnormal returns. For example, Jegadeesh and Titman (1993) use a CAPM benchmark, while Grundy and Martin use a conditional factor model with two factors (the market portfolio and a size factor). However, if these pricing models are misspecified, then the abnormal returns which follow from their use are misspecified as well. That is, as Fama (1998) points out, these studies are plagued by a “bad model” problem.

The method that we employ follows Chen and Knez (1996) in using a stochastic discount factor retrieved from a set of basis assets to measure risk-adjusted performance. The existence of a stochastic discount factor is a necessary and sufficient condition for equilibrium in securities markets; thus, evidence that a stochastic discount factor prices the momentum strategies is consistent with the ability of risk to explain these profits. However, unlike parametric performance measures, which adopt candidate stochastic discount factors implied by particular asset pricing models, nonparametric performance measures attempt to recover a set of admissible stochastic discount factors

based on minimal conditions such as the *law of one price* or *no arbitrage* conditions. Thus, relative to parametric approaches, this study asks whether momentum profits can be explained with the minimal restriction of equilibrium in securities markets. Our approach provides insight into whether the profitability of such strategies represents the failure of particular parametric models in describing the cross-section of returns, or the failure of the rational pricing paradigm.

This approach has several advantages. First, as mentioned above, estimating a stochastic discount factor from a set of basis assets imposes equilibrium pricing conditions without the need to specify a parametric benchmark. Second, the estimation of a discount factor leads to natural measures of risk-adjusted abnormal performance. If the trading strategies considered outperform when measured in this manner, then it is more likely that their performance is due to investor irrationality. However, if these strategies cannot outperform the benchmark, their success may be consistent with rational asset pricing. Third, the nonparametric measures we use can be easily extended to conditional measures which incorporate the possibility that risk premiums are time-varying. This is particularly important given the recent work by Chordia and Shivakumar (2000) who present evidence that momentum profits are related to business cycle conditions. Their model for expected returns, however, imposes no cross-sectional constraints. In our framework, we can test whether time-variation in expected returns might contribute to momentum “profits” while simultaneously requiring that those returns satisfy equilibrium constraints such as the law of one price.

As basis assets, we use industry-sorted portfolios. Moskowitz and Grinblatt (1999) also employ industry-sorted portfolios in their paper, but the focus of our work is very different. Specifically, Moskowitz and Grinblatt assume that industry components are “non-priced” risk factors, and test whether the use of industry information results in higher abnormal profits. Their “industry momentum” strategy is dynamic in nature, and consists of buying the top three winner and selling the bottom three loser industry portfolios. In contrast, we use all 20 industry portfolios to extract a pricing kernel and then test whether this pricing kernel can also explain individual security momentum profits (we separately and subsequently ask whether it can explain industry momentum profits.) In the unconditional setting, the weights on our industry portfolios are *static*; in the conditional setting, these weights vary *only* as a function of three public conditioning variables

and not as a function of the past performance of the industry portfolios. Put another way, our tests simply ask whether the momentum strategies add anything to the opportunity set (or mean-variance frontier) formed by the industry portfolios or the managed portfolios thereof.¹ A dynamic, actively managed scheme, such as that used in Moskowitz and Grinblatt (1999), is not included in the investor's opportunity set as we construct it; in a later section of the paper, we test whether the industry momentum strategy adds significantly to that opportunity set.

Our results suggest that, when investors do not condition on public information, approximately half of the momentum profits can be achieved by holding a fixed-weight combination of the set of basis assets, where the basis assets consist of industry-sorted portfolios. Specifically, eight (nine) out of the 16 strategies remain profitable at the 5% (10%) level when we require that the law of one price hold, and the average profit declines to 52% of the levels achieved before we allow for risk adjustment. Thus, the nonparametric risk-adjusted performance of the strategies averages half of that of the raw returns. In addition, a joint test of all 16 strategies fails to reject the null hypothesis of no abnormal performance. This contrasts with the results obtained when using a CAPM benchmark, in which 12 of the strategies have risk-adjusted performance higher than the raw profits, and 13 of the strategies examined remain significant at the 5% level after adjusting for market risk. These results also contrast sharply with those obtained using the Fama-French benchmark, in which all 16 of the strategies retain significant positive profits and, indeed, all 16 risk-adjusted profits are magnified relative to their raw levels. If we require that a stronger no-arbitrage condition hold, that is, require that the pricing kernel be non-negative and again do not allow conditioning, the results are similar. Average abnormal profits decline substantially, with seven of the 16 strategies' profits remaining unexplained at the 5% level. When we allow for investors' expectations to vary conditional on three publicly known conditioning variables, momentum strategy profits decline further, to approximately one-third of the level of raw profits obtained by assuming that the law of one price holds, and only one-fourth of the level of profits using the NA estimation technique. None of the 16 momentum strategies remain profitable at the 5% level using either estimation technique. These results are consistent with those of Chordia

¹A one-to-one correspondence exists between the Hansen-Jagannathan bounds and the mean-variance frontier when the law of one price is assumed. However, under the assumption of no arbitrage, the link between the bounds and the frontier is broken. Consequently, in the no arbitrage setting, our tests are more restrictive than a diagnostic implied by the mean-variance frontier.

and Shivakumar (2000), who suggest that time-variation in expected returns explains a portion of momentum strategy profits. Overall, we contend that, given the cross-sectional restrictions imposed by assuming the existence of a stochastic discount factor, these results support the ability of a rational pricing model to describe the returns to momentum strategies.²

These results are surprising in light of the large profits documented for winner and loser portfolios relative to parametric benchmarks. However, the nonparametric results that we present in the paper suggest that momentum strategies so dramatically outperform previously considered benchmarks not because of information contained in past prices, but rather because of the nature of the risk adjustment. We develop a simple theoretical model of the risk measures of securities in the extreme winner and loser portfolios, measured relative to the ‘true’ stochastic discount factor, and show that the risk measure of a momentum portfolio should be increasing in the risk premium. This contrasts sharply with the results in previous work examining momentum profits, which suggest that momentum portfolios are associated with negative risk measures. We present evidence that risk measures of the momentum portfolios in our sample, estimated relative to a value-weighted market portfolio, diverge significantly from this standard. In contrast, betas estimated relative to the alternative pricing kernels estimated in our paper conform more closely to the predicted behavior. Finally, we briefly discuss the interpretation of our results and their implication for pricing models of various types.

The remainder of the paper is outlined as follows. In Section 2, we discuss the theoretical basis for the existence of the measure of excess performance used in the paper, largely following Chen and Knez (1996), and outline the methods used for estimating the risk-adjusted performance of the strategies. In Section 3, we describe the data; estimation results are presented in Section 4. In Section 5, we formalize the theoretical link between past return information and risk, perform an empirical investigation of these implications and examine the properties of the test statistics using Monte Carlo techniques. We discuss the interpretation of the results in Section 6 and briefly

²Our contention relies on the notion that the conditional benchmark reflects the existence of equilibrium in an economy with time-varying risk premia. It is possible that instead the results suggest a type of feedback or momentum *between* markets in which equilibrium (based on the price responses to public information) is determined only with a lag. In Section 5, we provide evidence that suggest that our results may be driven by risk adjustment, rather than lagged equilibrium. However, we cannot rule out a market-wide (as opposed to asset-specific) delayed response using our techniques.

summarize in Section 7.

2 Nonparametric Performance Measures of Trading Strategy Portfolios

In this section, we discuss the implications of the existence of a stochastic discount factor for the analysis of the profitability of trading strategies.

2.1 Stochastic Discount Factors and Performance Measures

Under suitable regularity conditions, Harrison and Kreps (1979) prove that the absence of arbitrage opportunities in a securities market implies the existence of a stochastic discount factor. This stochastic discount factor satisfies the pricing relationship

$$E[m_{t+\tau} x_{i,t+\tau} | \mathcal{F}_t] = 1 \quad \forall \quad i, \tau. \quad (1)$$

where $x_{i,t+\tau} = 1 + R_{i,t+\tau}$ denotes the gross return on traded asset i in the economy, $\mathcal{F}_t$ is a filtration, representing information available to the investing public at time t , and $m_{t+\tau}$ denotes the economy-wide stochastic discount factor or pricing kernel. If we assume the existence of a riskfree asset, then for any excess return (1) implies

$$E[m_{t+\tau} r_{i,t+\tau} | \mathcal{F}_t] = 0 \quad (2)$$

where $r_{i,t+\tau} = R_{i,t+\tau} - R_{f,t}$ denotes the excess return on asset i in the economy. The returns to the trading strategies discussed in the introduction, which buy winning securities and sell losing securities, are of the form of equation (2). Under the law of one price, an $m_{t+\tau}$ should exist that prices the gross (and excess) returns on these securities as well. Under the stronger condition of no arbitrage, this $m_{t+\tau}$ must also be positive.³

Intuitively, this formulation states that the ‘fair’ or risk-adjusted return of any asset (or trading strategy) will be determined by the pricing kernel or stochastic discount factor in the economy. Note that this formulation also expressly accounts for risk. That is, (2) can be rewritten as:

$$E[r_{t+\tau}] = -\frac{1}{E[m_{t+\tau}]} \text{Cov}[r_{t+\tau}, m_{t+\tau}] \quad (3)$$

³ Following Hansen and Jagannathan (1991), we use a non-negativity condition instead of the positivity condition in actual estimation.

The covariance of the return on the asset with the pricing kernel varies for each asset, and reflects its risk: if an asset's payoffs are greater in good states (when the value of m is lower), its expected risk premium is higher.

Hansen and Jagannathan (1991) investigate how to retrieve the stochastic discount factors $m_{t+\tau}$ from a given set of tradable, or basis, assets. The key underlying assumption therein is that there is no pricing inconsistency among the basis assets: i.e., the stochastic discount factors are *admissible*. Hansen and Jagannathan (1991) suggest two particular solutions for $m_{t+\tau}$ which are the minimum-norm discount factors defined in different metrics. The first is defined as the $m_{t+\tau}$ that satisfies equations (1) (or equivalently (2)) and is in the linear span. That is

$$m_{t+\tau}^{LOP} = \mathbf{x}'_{t+\tau} \delta \quad (4)$$

This stochastic discount factor has been studied extensively in the literature [e.g. Chamberlain and Rothschild (1983), Hansen and Jagannathan (1991, 1997)]. We follow Chen and Knez (1996) and term this solution for the stochastic discount factor the law of one price (LOP) discount factor, since its existence necessitates only that the law of one price hold.

The second stochastic discount factor satisfies equation (1) and the further requirement that $m_{t+\tau}$ is non-negative, or

$$m_{t+\tau}^{NA} = (\mathbf{x}'_{t+\tau} \delta)^+ \quad (5)$$

where $(\mathbf{x}'_{t+\tau} \delta)^+ \equiv \max(\mathbf{x}'_{t+\tau} \delta, 0)$. This stochastic discount factor satisfies the stronger condition of no arbitrage, thereby ruling out investment opportunities with positive payoffs and non-positive prices. We refer to this discount factor as the no arbitrage (or NA) stochastic discount factor.

The implications of relationships (1) and (2) for trading strategies are straightforward. Consider the set of basis assets from which the strategies are to be implemented, $\mathbf{x}_{t+\tau}$. Under the law of one price (or, alternatively no arbitrage), there must be a stochastic discount factor that correctly prices $\mathbf{x}_{t+\tau}$ and all linear combinations thereof. Furthermore, this must hold even in the case in which the combinations represent zero-cost portfolios. Thus, the testable implication of equations (1) and (2) for trading strategies is that

$$\begin{aligned} & \exists \ m^* \text{ s.t.} \\ & E[m_{t+\tau}^* \mathbf{x}_{B,t+\tau} | \mathcal{F}_t] = \mathbf{1}_N \\ & E[m_{t+\tau}^* r_{TS,t+\tau} | \mathcal{F}_t] = 0 \end{aligned} \quad (6)$$

where $\mathbf{x}_{B,t+\tau}$ represents the gross returns on the basis assets over τ periods and $r_{TS,t+\tau}$ represents the return to the trading strategy over τ periods.

Chen and Knez (1996) show that relationship (6) provides the basis for a test of the risk-adjusted performance of a portfolio. According to (6),

$$\alpha_\tau = E[\alpha_{t,\tau}] = E[m_{t+\tau}^* r_{TS,t+\tau}] = 0 \quad (7)$$

Therefore, a natural hypothesis to test is whether α_τ is equal to zero. This measure is similar to other abnormal performance measures such as Jensen's (1968) alpha, which obtains as a special case of (7). However, unlike the Jensen measure, (7) does not rely on the existence of a particular model of equilibrium asset prices.

It is important to note the differences between this method and other approaches to assessing the abnormal performance of momentum strategies. Other researchers assume a particular parametric pricing model which is assumed to price all assets, then test whether the momentum strategy outperforms relative to this model. For example, Jegadeesh and Titman (1993) employ a CAPM benchmark, thereby using the pricing kernel

$$m_{t+1} = \phi_0 + \phi_1 R_{M,t+1},$$

where $R_{M,t+1}$ is the return on a particular market-proxy portfolio. Grundy and Martin (1999) employ a conditional factor model, with the market portfolio and a size factor representing the set of priced factors. In their results, these researchers find evidence that risk-adjusted returns of momentum strategies are significant, and frequently larger than the raw returns themselves. Thus, *conditional on a particular parametric pricing model being true*, their results suggest that momentum strategies generate statistically and economically significant profits and in fact may in some cases have negative risk measures. Given the lack of consensus in the literature about which pricing model is more "correct" (and, in fact, the large body of literature that suggests that the CAPM is not empirically supported in the data), it seems worthwhile to examine the abnormal performance of such strategies without the need to specify a particular parametric model. Rather, the model we employ assumes first, that the set of basis assets chosen form a basis of comparison for the remaining assets in the economy, and second, that the pricing kernel can be accurately retrieved from this set of assets. Given such a pricing kernel, we test whether the momentum strategy enhances the investor's opportunity set, or whether the momentum profits can be achieved

through a combination of the basis assets. We provide some additional comparisons between the pricing kernel we extract and other parametric benchmarks in Section 5.

2.2 Methods

Estimating α and testing (7) is straightforward and can be accomplished easily via Hansen's (1982) Generalized Method of Moments (GMM). In order to implement this procedure, we substitute $\mathbf{x}'_{t+\tau}\delta ((\mathbf{x}'_{t+\tau}\delta)^+)$ for $m_{t+\tau}^{LOP}(m_{t+\tau}^{NA})$ and collect the vector of errors

$$\epsilon_{t+\tau} = (\mathbf{u}'_{t+\tau}, v_{t+\tau})'$$

where

$$\begin{aligned}\mathbf{u}_{t+\tau} &= \mathbf{x}_{B,t+\tau}\mathbf{x}'_{B,t+\tau}\delta - \mathbf{1}_n \\ v_{t+\tau} &= r_{TS,t+\tau}\mathbf{x}'_{B,t+\tau}\delta.\end{aligned}$$

We then form the vector of sample moment conditions

$$\mathbf{g}_T(\delta) = \frac{1}{T} \sum_{t=1}^T \epsilon_{t+\tau}$$

Under the null hypothesis of no abnormal performance, (7), $E[\mathbf{g}_T(\delta)] = 0$. We test this hypothesis by minimizing the quadratic form

$$J_T = T\mathbf{g}'_T(\delta)\mathbf{W}_T\mathbf{g}'_T(\delta)$$

where

$$\mathbf{W}_T = E[\mathbf{g}_T(\delta)\mathbf{g}'_T(\delta)]^{-1}$$

As shown by Hansen, under the null $J_T \sim \chi^2_{n-k}$, where n is the number of moment conditions and k is the number of parameters. Each of the n assets is associated with one moment condition. We incorporate the return on a risk-free asset into the estimation as well; Dahlquist and Soderlind (1999) emphasize the importance of this technique in order to fix the mean of the stochastic discount factor at a reasonable level. The trading strategy return adds an additional moment condition so that $J_T \sim \chi^2_1$.

We have limited our discussion thus far to agents employing static trading strategies in the formulation of the basis portfolios, i.e., buy-and-hold strategies. This approach is equivalent to

asking whether the momentum strategy returns are spanned by a constant-weight, linear combination of the basis assets, or a constant risk premium. This static approach ignores a large body of evidence that suggests that expectations vary over time.⁴ In fact, Chordia and Shivakumar (2000) argue that such time-variation in expected returns can fully explain momentum strategy profits (although as we mention above, they examine expected returns in a setting which does not impose equilibrium constraints.) Consequently, we estimate an alternative pricing kernel, allowing for the use of a limited set of conditioning information. The conditioning information that we use does not rely explicitly (or only) on information contained in past prices; rather, it takes the form of public information known to economic agents at the time of portfolio formation. As noted by Fama (1991), this type of conditioning does not violate market efficiency if risk premia are time-varying. Intuitively, allowing for the use of conditioning information expands the set of basis assets to include managed portfolios, which take into account changes in the moments of returns that are related (only) to changes in the conditioning variables. (See also Cochrane (1997) for an excellent discussion of the nature of conditional tests.)⁵

In order to incorporate the idea of time-varying expectations, we follow Hansen and Singleton (1982) in assuming that conditional expectations are linear in time t information variables, $\mathbf{Z}_t$. This case yields the following representation for the pricing relation (1)

$$E[(\mathbf{x}_{t+\tau} \otimes \mathbf{Z}_t)(\mathbf{x}_{t+\tau} \otimes \mathbf{Z}_t)'\delta] = \mathbf{1}_N$$

We can then collect the vector of errors

$$\epsilon_{z,t+\tau} = (\mathbf{u}'_{z,t+\tau}, v_{z,t+\tau})'$$

where

$$\mathbf{u}_{z,t+\tau} = (\mathbf{x}_{t+\tau} \otimes \mathbf{Z}_t)(\mathbf{x}_{t+\tau} \otimes \mathbf{Z}_t)'\delta - \mathbf{1} \otimes \mathbf{Z}_t$$

⁴ The literature documenting time variation in expected returns is voluminous. Some articles include Campbell (1987), Fama and French (1988), and Shanken (1990). The implications of this time variation for asset pricing models has been investigated theoretically in Hansen and Richard (1987) and empirically in Gibbons and Ferson (1985), Harvey (1989), and Ferson and Harvey (1991), among others.

⁵ Another implicit assumption in our use of conditioning information is that we are ruling out mispricing not only in the basis assets, but also in the conditioning variables. Thus, we assume that the ability of conditioning variables to predict returns is not due to an aggregate lagged price adjustment. However, this assumption is also central to any equilibrium parametric model of risk adjustment.

$$\mathbf{v}_{z,t+\tau} = r_{TS,t+\tau}(\mathbf{x}_{t+\tau} \otimes \mathbf{Z}_t)' \delta$$

The no-arbitrage discount factor can also be employed in the methods discussed above. It requires that we replace $\mathbf{x}'\delta$ with $(\mathbf{x}'\delta)^+ \equiv \max(\mathbf{x}'\delta, 0)$ in the unconditional setting and $(\mathbf{x} \otimes \mathbf{Z})'\delta$ with $((\mathbf{x} \otimes \mathbf{Z})'\delta)^+$ in the conditional setting. The estimation of this quantity is significantly more complicated, but can be achieved through the procedure in Hansen and Jagannathan (1991). This approach also imposes the restriction that the estimated stochastic discount factor has the minimum second moment in the set of positive pricing kernels.

3 Data

As a starting point, we consider the universe of firms listed on CRSP over the period 12/31/62 through 12/31/97, yielding 420 monthly observations. We construct the momentum payoffs and the basis portfolios as follows.

3.1 Momentum Portfolios

We follow the method outlined in Jegadeesh and Titman (1993) for the construction of the momentum strategy payoffs. NYSE and AMEX firms are classified into ten deciles on the basis of the past $J = \{3, 6, 9, 12\}$ month cumulative holding period returns at month t . The firms offering the best performance over these horizons are termed “winner” firms and the firms offering the worst performance over these horizons are termed “loser” firms. The winner and loser firms are then held in portfolios for the next $K = \{3, 6, 9, 12\}$ months. That is, at time t for horizon J , we purchase an equally-weighted portfolio of firms that were winners from $t - J$ through t and sell an equally-weighted portfolio of firms that were losers over period $t - J$ through t . Since the $J = 12, K = 12$ strategy requires 24 months to implement, this reduces the number of time series observations to 396, covering the period 1/31/65 through 12/31/97.

3.2 The Basis Assets

An important consideration in estimating the measure of abnormal performance discussed in the paper is the choice of reference assets. In complete markets, the stochastic discount factor will be unique. However, when markets are incomplete, there exists a multiplicity of stochastic discount

factors that will correctly price the assets in the economy [Harrison and Kreps (1979)]. If the reference set from which the stochastic discount factor is formed spans the payoff opportunity set which is available to investors, then measuring abnormal performance relative to this reference set will provide a correct (and unique) inference. However, if the reference set does not span the payoffs, it is possible to incorrectly reject the null hypothesis of zero abnormal performance.⁶ In Section 5, we will discuss whether the retrieved stochastic discount factors are admissible in the context of a specific analysis of momentum trading strategies.

Ideally, to prevent an incorrect rejection of the null, the reference assets should mimic the entire opportunity set from which the trading strategies are chosen. However, since the momentum strategies we investigate are formed from the set of all NYSE and AMEX securities, this approach cannot be implemented in practice. Therefore, we must choose a more parsimonious set of reference assets that capture as much of the investment opportunity set as possible, that is, we wish to group securities in a manner that maximizes intra-group correlation and minimizes inter-group correlations. King (1966) demonstrates that industry groupings do precisely that. In an exhaustive analysis of factors important in the determination of stock returns, he concludes that market and industry factors capture most, if not all, of the common variation in stock returns. For example, he demonstrates that ‘large’ positive covariances in returns cluster strongly within industry groupings, and negative covariances are observed exclusively across industry groupings. Therefore, we form the reference set by forming portfolios on the basis of industry. Specifically, each year, we form 20 equally weighted portfolios of NYSE and AMEX firms on the basis of 2-digit SIC code groupings. Definitions of the groups are found in Table 2.⁷

Moskowitz and Grinblatt (2000) implement a variant of the momentum strategy which also relies on the use of industry portfolios. Specifically, they examine a strategy that buys the top three industry winners and sells the bottom three industry losers each period; they control for specific sources of risk (in their case given by the three Fama-French portfolios) to measure abnormal performance. They argue that this industry momentum strategy captures most of the momentum in

⁶Ahn and Shivdasani (1999) discuss this issue more fully and present methods for addressing the spanning problem.

⁷ We have also used industry portfolios reformed each month. The explanatory power of the basis assets is improved in that case. Note that Chen and Knez (1996) also use industry portfolios in their analysis of the performance of mutual funds.

individual securities. Moskowitz and Grinblatt’s work is quite different from our analysis. First, as in Jegadeesh and Titman (1993) and Grundy and Martin (1999), Moskowitz and Grinblatt choose a particular parametric risk adjustment, whereas we assume that industry portfolios can be rationally priced by some pricing kernel $m_{t+\tau}$ that may be represented by a single or multi-factor linear or nonlinear model. The purpose of the industry portfolios in this paper is merely to retrieve this kernel rather than to use industry-specific return components to ‘fine-tune’ a momentum strategy, or examine the sources of individual securities’ momentum profits. Second, the weights assigned to our industry portfolios are fixed (in the unconditional analysis) or are allowed to vary only with the three public information variables in the conditional tests, instead of the relative past performance of industry portfolios. That is, Moskowitz and Grinblatt (1999) examine the relation between two active trading strategies: an individual security momentum strategy and an industry momentum strategy. In our unconditional (conditional) tests, we are asking whether momentum strategy profits can be priced by passive combinations of industry (industry plus a particular set of managed) portfolios. These two schemes will coincide only if the industry momentum strategy is a reflection of the publicly available information we use in our tests. Third, we employ an (annually rebalanced) equal weighting in our industry portfolios, while Moskowitz and Grinblatt (1999) use a value weighting.⁸

3.3 Conditioning Information

Our conditional tests should in theory include all information relevant for investors to make portfolio decisions, as noted by Hansen and Richard (1987). In practice, we can only observe a subset of this information. Furthermore, econometric limitations suggest that in GMM estimation, power decreases as the number of moment conditions increases. Therefore, we consider a limited set of

⁸ We examined the possibility of using value weights in our industry portfolios. However, these portfolios performed poorly as basis assets, perhaps due to a lack of diversification in some portfolios. In our sample, we observed individual security weights as high as 60% in some months; this led to a large standard deviation of returns in these portfolios and a Hansen-Jagannathan bound much lower than that observed for equally weighted portfolios. That is, the equally-weighted portfolios capture wider cross-sectional variation in asset returns and thus impose a more informative restriction on the admissible set of stochastic discount factors. Consequently, we use equal-weighted portfolios in our analysis, but rebalance only annually to ensure that the transactions costs of holding the basis assets is not prohibitive.

information variables whose predictive power has been documented in the literature. In particular, we consider three information variables: the dividend yield on the S&P 500 index, the return on the Treasury Bill with maturity closest to one month, and the term spread, which is measured as the difference in yields on 10-year maturity Treasury Bonds over 1-year maturity Treasury Bills. The power of the first instrumental variable has been demonstrated by several authors, including Fama and French (1988). The second instrumental variable has been shown to have predictive power in many studies, including Fama and Schwert (1977) and Ferson (1989) and the third instrumental variable has been shown to predict stock returns by Fama and French (1988). The dividend yield data are obtained from Datastream International, the T-Bill return is taken from the Fama-Bliss risk-free rate file on CRSP, and the term spread data are obtained from the Federal Reserve report H.15. The dividend yield data are available from Datastream starting in January, 1965. Consequently, our sample period is reduced by one month to 395 observations.⁹

4 Estimation Results

4.1 Returns to Relative Strength Portfolios

Summary statistics for the returns to the momentum trading strategies are presented in Table 1. Consistent with Jegadeesh and Titman (1993), the mean monthly returns for the strategies vary widely, from 6 basis points per month for the 3-month/6-month strategy to 116 basis points per month for the 12-month/3-month strategy. As in Jegadeesh and Titman, the returns to the strategies tend to increase in the length of the portfolio formation period. However, the average returns to the strategies are almost always lower than those reported by Jegadeesh and Titman. At the extreme, the 12-month/3-month strategy reported in this paper has a mean return of 116 basis points per month compared to a return of 131 basis points per month reported by Jegadeesh and Titman. Much of this discrepancy is attributable to the sample period; the data in this paper cover an additional eight years of returns. Over a common sample period, the difference in mean returns is negligible.

In untabulated results, we investigate the ability of the conditioning variables to serve as in-

⁹ These three variables are similar to three (out of four) variables used in Chordia and Shivakumar (2000); they add a default spread variable as well.

struments for returns by regressing the industry portfolio returns on lagged values of the three variables. For all of the trading strategies, the lagged T-bill return is negatively related to the strategies' returns while the lagged dividend yield is positively related to the future returns. The coefficient estimates for these two conditioning variables are generally significant at the 5% level, indicating that they are informative instruments. The term-spread variable has poor explanatory power, with only two of 20 coefficients significant at the 5% level. The overall ability of the instruments to capture variation in the returns to the trading strategies is fairly good; the average R^2 is 0.0375, with a minimum value of 0.0150.

Table 2 summarizes monthly mean and standard deviation of returns on the twenty industry-sorted portfolios we form. The mean returns range from 101 basis points per month for utilities to 155 basis points per month for electrical equipment. The cross-sectional dispersion in mean returns is quite large. A similar pattern of variation in standard deviation in returns is documented in Panel B of Table 2.

Figures 1 and 2 display the Hansen and Jagannathan (1991) mean-standard deviation bounds on the set of admissible pricing kernels for the industry-sorted portfolios and the industry portfolios augmented by the trading strategies. Figure 1 presents unconditional bounds and Figure 2 presents conditional bounds on the kernels. The graphs present an informal measure of the industry portfolios' ability to span the trading strategies. In any analysis, when more assets are included the Hansen-Jagannathan bound either stays the same (a degenerate case) or shifts up. As noted by Ahn and Shivdasani (1999) [see also Snow (1992), DeSantis (1995), and Bekaert and Urias (1996)], there are two possibilities which drive the upward shift in the Hansen-Jagannathan bound. The first source of shift is spanning enhancement. That is, the basis assets chosen is not a good reference set, and the momentum portfolio captures factors which are absent in the determination of values of the basis assets (in this case, the industry portfolios). The second possibility is that the upward shift in the Hansen-Jagannathan bound arises as a result of mispricing of the momentum strategy portfolios. In both cases, we would reject the null hypothesis of no abnormal performance for the momentum portfolio, although in the first case the cause is a poor selection of reference assets. If the upward shift in the Hansen-Jagannathan bound is trivial, we can conclude that the momentum portfolios are well spanned by the basis assets and there are no pricing inconsistencies between the basis assets and the momentum strategy portfolios.

Figure 1 shows that the unconditional Hansen-Jagannathan bounds for the industry portfolios conforms fairly well to the bounds for the industry portfolios augmented by the strategy returns for most of the holding periods. Consistent with the returns observed in Table 1, the largest shifts occur in the 3-month holding period returns and the smallest in the 12-month holding period returns. Moreover, the graphs in Figure 2 suggest that the conditional Hansen-Jagannathan bounds for the industry portfolios and the industry portfolios augmented by the strategy returns are virtually indistinguishable. The bounds display no noticeable shift across the holding periods or the portfolio formation periods. Since, as shown in Hansen and Jagannathan (1991), the Hansen-Jagannathan bound is a mirror image of the mean-variance frontier, the absence of an upward shift in this bound is equivalent to the result that the mean-variance frontier faced by the investor does not shift to the left when momentum strategies are included in the opportunity set.¹⁰ Thus, the graphs suggest the conclusion that the industry portfolios span the trading strategies, particularly in the conditional setting, and the momentum portfolios do not appear to enhance the mean-variance frontier. We formally test these inferences in the next section.

4.2 Measuring Abnormal Performance

As noted in the previous subsection, the returns to the strategies are fairly large and vary across both formation and holding periods. In this subsection, we assess the degree to which risk can explain this variation.

4.2.1 CAPM-Based Performance Measure

Table 3 presents an analysis of the abnormal performance of the trading strategies using a CAPM benchmark for risk. This analysis follows that presented in Jegadeesh and Titman (1993) and consists of regressing the returns to the trading strategies on the excess market return and a constant

$$r_{TS,t+\tau} = \alpha_{TS} + \beta_{TS}(R_{M,t+\tau} - R_{f,t+\tau}) + \epsilon_{TS,t+\tau}$$

The trading strategy earns an abnormal return if $\alpha_{TS} \neq 0$.

¹⁰This equivalence holds for the LOP Hansen-Jagannathan bound but not for the NA Hansen-Jagannathan bound. In the NA setting, the link between the mean-variance frontier and the bound is broken; an upward shift in the NA bound does not have the same implication for the mean-variance frontier as a shift in the LOP bound.

The results of Table 3 suggest that 13 of the 16 trading strategies earn positive abnormal returns. At the extremes, the 3-month/6-month strategy earns the smallest abnormal return, 15 basis points per month, whereas the 12-month/3-month strategy earns the largest, 123 basis points per month. Jegadeesh and Titman (1993) present results only for the 6-month/6-month strategy and find that the strategy earns an abnormal return of 100 basis points per month, which is slightly higher than the abnormal return of 81 basis points per month shown in Table 3. However, this differential is proportional to the difference in raw returns discussed in the previous subsection.

Under the assumption that the CAPM is the relevant pricing model and that risk measures are well-estimated, the conclusion drawn from Table 3 is that a momentum trading strategy provides significant excess returns on a risk-adjusted basis. This evidence is consistent with that in Jegadeesh and Titman (1993), who report that momentum profits persist even after CAPM risk adjustments. However, given the large body of evidence which suggests that the CAPM does not hold [e.g, Fama and French (1992) or MacKinlay (1986)], we explore alternative benchmarks.

4.2.2 Fama-French Performance Measure

Table 4 presents abnormal performance measures of momentum trading strategies using the Fama-French (1992) three factor model to adjust for risk. Thus, the following model is estimated:

$$r_{TS,t+\tau} = \alpha_{TS} + \beta_{TS,MRP}(R_{MRP,t+\tau}) + \beta_{TS,SMB}(R_{SMB,t+\tau}) + \beta_{TS,HML}(R_{HML,t+\tau}) + \epsilon_{TS,t+\tau},$$

where R_{MRP} is the return on a market proxy, R_{SMB} is the return on a zero-cost portfolio that buys large capitalization firms and sells small capitalization firms and R_{HML} is the return on another zero-cost portfolio that buys high book to market firms and sells low book to market firms. As before, we test whether the intercept is significantly different from zero. The results in Table 4 show that, using this model to adjust for risk, 16 out of 16 trading strategies earn positive abnormal returns, with very small associated p-values. As with the CAPM risk adjustment, the largest abnormal return is in the 12-month/3-month strategy with an abnormal return of 1.69% per month. This evidence is consistent with the results presented in Fama and French (1996) and Carhart (1997), who show that the three factor model does not explain momentum profits.

4.2.3 Unconditional Performance Measures

Next, we examine the implications of the unconditional performance measure based on the law of one price. As discussed previously, the basis used for the tests consists of the 20 industry-sorted portfolios augmented by the return on the riskless asset. Results of this estimation are shown in Table 5.

The results in Table 5 suggest that the abnormal performance of momentum strategies declines substantially when the unconditional performance measure is used. Of the 16 individual strategies analyzed, eight continue to exhibit abnormal performance at the 5% level. However, even for these eight strategies, the abnormal profits fall on average by 39 basis points or an average of 43% relative to the raw profits. In one additional case, the abnormal performance is significant at the 10% level. The average profit level is 52% of the raw profits observed in Table 1. A joint test across all 16 momentum strategies fails to reject the null hypothesis of no abnormal performance, with a p-value of 0.463. This evidence suggests that it is possible to construct a stochastic discount factor which prices the industry-sorted portfolios, as well as encompassing approximately half of the profits of the momentum portfolios we form.

In Table 6, we require that the stronger no-arbitrage condition hold, and again investigate whether a stochastic discount factor formed from the basis assets can explain the momentum profits observed in the data. The results are similar: seven of the sixteen strategies retain profits which are significant at the 5% level and an additional two have profits significant at the 10% level. On average, abnormal profits still decline substantially, to average 49% of the levels observed in Table 1. In addition, a joint test across all 16 momentum strategies fails to reject the null hypothesis of no abnormal performance with a p-value of 0.469.

Overall, the evidence in Tables 5 and 6 suggest that, while a specific factor model (i.e., the CAPM) cannot explain momentum profits (and in fact appears to magnify these profits), a pricing kernel estimated from industry-sorted portfolios, which satisfies the much weaker condition of the law of one price and which is stationary over time, can explain a substantial fraction of these profits. Restricting the pricing kernel to the positive range provides similar results—the abnormal profits of the momentum strategies we consider decline by about half.

4.2.4 Conditional Performance Measures

Results of implementation of the conditional LOP performance measure are presented in Table 7. When we allow investors to rebalance their portfolios by conditioning on public information, we find no evidence for abnormal profits in momentum portfolios. Of the 16 trading strategies, none earns a significant abnormal return at the 5% level; five are significant at the 10% level. The strategies earn, on average, 33% of the raw profits observed in Table 1. Two strategies earn negative profits. The highest abnormal performance is still observed in the 12-month/3-month strategy, at 43 basis points per month. The abnormal performance of the 6-month/6-month strategy investigated in detail in Jegadeesh and Titman (1993) falls from 85 to 31 basis points per month. Not surprisingly, the joint test again fails to reject the null of zero abnormal performance ($p\text{-value} = 0.592$).

The results with the no-arbitrage measure presented in Table 8 are similar: only two strategies are significant at the 10% level. Moreover, the joint test fails to reject zero abnormal performance ($p\text{-value} = 0.380$). Overall, this evidence suggests that momentum profits do not persist when investors are allowed to consider the effects of public information on the investment opportunity set. That is, if we allow for time-varying expected returns, the profitability of the momentum strategy portfolio can be construed as a reward for the risk inherent in these portfolios.

These results are consistent with those of Chordia and Shivakumar (2000) who find that momentum profits vary strongly with the business cycle and a four-factor time-varying model of expected returns appears to subsume much of the profits associated with momentum portfolios. In our tests, however, any time-varying expected returns in the basis assets must still satisfy the equilibrium requirement of the law of one price (or no-arbitrage). Of course, if markets are not in equilibrium (or take an extended period to achieve equilibrium), then our measure of abnormal performance may be mis-specified. For example, if the predictability in returns for which we find evidence is due to a lagged price response to public information in the basis assets, rather than time-variation in risk premia, then our measure of abnormal performance may be mis-specified. This violation of our equilibrium condition, however, would affect not only our nonparametric tests but any equilibrium parametric model as well.¹¹

¹¹ That is, assuming that the parameters of a particular pricing model are estimated from a set of basis assets, inferences based on such a model would contain two possible sources of error: the first from any mispricing in the basis assets, and the second from potential mis-specification of the

5 An Analysis of Risk Adjustment

Coupled with the evidence in the diagnostic analysis of the Hansen-Jagannathan bound, the empirical results in the previous section point toward the conclusion that momentum strategies do not outperform other asset combinations through time. Given that our results are consistent with economic risk premiums as the source of the profitability of trading strategies, why, then, have the existing studies based on specific parametric benchmarks failed to detect this? In this section, we formalize the link between past return information and risk to answer this question. This study also provides us with an additional diagnostic tool for evaluating whether the benchmark stochastic discount factor proposed is *admissible* in the context of measuring the (risk-adjusted) profitability of the trading strategies.

We assume the fundamental valuation equation, (1), as in Section 2. If a risk-free asset exists, we can rewrite this expression (time subscripts suppressed for convenience) as

$$E[R_i] - R_f = -\frac{1}{1 + R_f} \text{Cov}(m, R_i) \quad (8)$$

Given this relation, it is well-known (see, e.g., Cochrane (1997)) that one can construct a hedge portfolio or constant consumption portfolio which is perfectly correlated with the stochastic discount factor, m . Then, if we denote the return on such a hedge portfolio as R_d , equation (8) can be further simplified to

$$E[R_i] - R_f = \beta_{di} (E[R_d] - R_f), \quad (9)$$

The above expression is similar to the CAPM, but the hedge portfolio does not need to be a market portfolio; in addition, this expression is still consistent with a multi-factor model.¹² In this characterization, β_{di} is similar to the market beta, and measures the sensitivity of the asset's payoff to that of the constant consumption portfolio, R_d .

Given equation (9), the realized return on asset i can be expressed as a function of its beta and the excess return on the hedge portfolio. Let λ_t be the time t realized excess return on the hedge pricing model.

¹² The existence of multiple factors in any form can be nested since (8) is a mathematical identity to (9). The difference between the two equations is similar to the difference in the representation of expected returns between the Consumption CAPM of Breeden (1979) and the Intertemporal CAPM of Merton (1973). The multiple factors are reduced to a single stochastic variable which represents the intertemporal marginal rate of substitution. In our context, the pricing kernel m and the return on its corresponding hedge portfolio, R_d , can be a function of multiple factors.

portfolio, $R_d - R_f$. We assume that $\lambda_t \sim N(\bar{\lambda}, \sigma_\lambda^2)$. Then the realized return on asset i at time t can be represented as:

$$\begin{aligned} R_{it} - R_f &= \beta_{di} \lambda_t + \epsilon_{it} \\ &= \beta_{di} \bar{\lambda} + \beta_{di} (\lambda_t - \bar{\lambda}) + \epsilon_{it}, \end{aligned}$$

where $\epsilon_{it} \perp \lambda_t$.

5.1 The Risk Characteristics of Extreme Performers

In this subsection, we analyze the relationship between the risk of an asset, β_{di} , and the probability of the security being selected as ‘extreme’. The theoretical relationship we derive is a direct outcome of the fundamental valuation equation (8), which holds in the absence of arbitrage opportunities. As such, it can serve as a diagnostic tool for evaluation of the admissibility of the stochastic discount factor proposed, and against which the performance of trading strategies is measured.

To compare potential candidates for benchmark models, we first examine the portfolios of extreme performers in order to determine if there are predictable patterns in their risk characteristics. For convenience, assume that there are ten assets under consideration; these assets can be characterized by the magnitude of their respective betas, such that¹³

$$\beta = (\beta_{d1} \ \beta_{d2} \ \cdots \ \beta_{d9} \ \beta_{d10})',$$

where $\beta_{d1} < \beta_{d2} < \cdots < \beta_{d9} < \beta_{d10}$. Further, without loss of generality we assume that $\text{var}(\epsilon_{it}) = \sigma_\epsilon^2 \ \forall \ i = 1, \dots, 10$. With these assumptions, we can describe the joint distribution of asset returns as:

$$R - R_f \sim \text{MVN}_{10}(\beta \bar{\lambda}, \ \beta \beta' \sigma_\lambda^2 + \sigma_\epsilon^2 I),$$

where the volatility of the risk premium, σ_λ , is equal to σ_{Rd} and I denotes the identity matrix.

We define the probabilities of a particular firm being selected as a winner or loser as:

$$\begin{aligned} Pr_i^h &= \text{Probability} [\text{asset } i \in W] \\ Pr_i^l &= \text{Probability} [\text{asset } i \in L]. \end{aligned}$$

¹³ We assume 10 assets as an analogy to a momentum investment strategy which takes a long position on firms in the top decile (winner) of past performance and a short position on the bottom decile (loser).

Clearly, the probability of a firm being selected as ‘extreme’ will depend both on the firm’s return, and the threshold used to characterize a security as extreme. For simplicity, we focus on the securities which are designated as extreme winners. A similar analysis can be performed for securities characterized as ‘losers’.

Let $\delta_t^w(\lambda_t)$ denote the threshold value which characterizes an asset as a ‘winner’. This threshold is a function of a particular realization of the excess return on the hedge portfolio d . Given the assumptions and definitions above,

$$Pr_i^h = \text{Probability} [\beta_{di}\lambda_t + \epsilon_{it} \geq \delta_t^w(\lambda_t)],$$

where

$$\begin{aligned} \delta_t^w(\lambda_t) &= \max(R_{1t} \ R_{2t} \ \cdots \ R_{9t} \ R_{10t}) \\ &= \max(\beta_{d1}\lambda_t + \epsilon_{1t} \ \beta_{d2}\lambda_t + \epsilon_{2t} \ \cdots \ \beta_{d9}\lambda_t + \epsilon_{9t} \ \beta_{d10}\lambda_t + \epsilon_{10t}). \end{aligned}$$

Suppose that $\sigma_\epsilon = 0$, i.e., there is no unsystematic risk. Then it is clear that

$$\delta_t^w(\lambda_t) = \delta_t^{w0} = \begin{cases} \beta_{d10}\lambda_t & \text{if } \lambda_t \geq 0 \\ \beta_{d1}\lambda_t & \text{if } \lambda_t < 0 \end{cases}$$

The intuition behind this result is straightforward. With no idiosyncratic risk, only the highest and the lowest beta assets can be ‘winners’. When the constant-consumption portfolio return moves in a positive direction, the highest beta asset will be the winner. In contrast, when the market reverses, the lowest beta asset will be the winner with probability one.

In the more relevant case, where $\sigma_\epsilon \neq 0$, then the threshold values used to characterize securities as extreme will be affected as,

$$\delta_t^w(\lambda_t) = \delta_t^{w0} + f(\beta, \lambda_t, \sigma_\epsilon),$$

where $f(\cdot)$ has a second order effect provided the number of groupings is small relative to the number of securities.¹⁴

Based on this, the probability of a particular security being categorized as a winner still depends on the security’s relative beta. However, in addition to this effect, the probability of being placed in the winner category also depends on the realization of the unexpected portion of its return:

$$Pr_i^h \approx \text{Probability} [\epsilon_{it} > \eta_i(\text{sign}(\lambda_t))\lambda_t], \quad (10)$$

¹⁴ Although we could improve this approximation by elaborating the functional form of $f(\cdot)$ using a polynomial series, δ_t^{w0} is enough for a qualitative analysis of the relationship between the probability that a firm is a ‘winner’ and its beta.

where

$$\eta_i(\text{sign}(\lambda_t)) = \begin{cases} \beta_{d10} - \beta_{di} & \text{if } \lambda_t \geq 0 \\ \beta_{d1} - \beta_{di} & \text{if } \lambda_t < 0 \end{cases}.$$

In Appendix 1, we investigate this analytical expression for the probability of a security with a particular β_{di} being chosen in the extreme portfolio; Figure 3A illustrates the performance of the approximation in equation (10), using simulated security returns grouped into ten portfolios. The approximation appears to be reasonable. Therefore, we use this expression to evaluate the relationship between the probability of a security being deemed a ‘winner’ and its β_{di} . Specifically, we show that the probability of being a winner when $\lambda_t \geq 0$ ($\lambda_t < 0$) is an increasing (decreasing) function of β_{di} .

Lemma 1: *The probability that an asset will be a ‘winner’ (‘loser’) in a positive market is an increasing (decreasing) function of the asset’s β_{di} . In contrast, the probability that an asset will be a ‘winner’ (loser) in the reverse market is a decreasing (increasing) function of β_{di} : i.e.,*

$$\begin{aligned} \frac{\partial \text{Prob}(r_{it} \in H | \lambda_t \geq 0)}{\partial \beta_i} &> 0 \\ \frac{\partial \text{Prob}(r_{it} \in H | \lambda_t < 0)}{\partial \beta_i} &< 0. \end{aligned}$$

Proof: See Appendix 1.

Lemma 1 states that even when $\sigma_\epsilon \neq 0$, the intuition drawn from the case of $\sigma_\epsilon = 0$ still holds, even though any asset now has some positive probability of being selected as a winner. One important implication of this lemma is that there may not be a monotonic association between a firm’s beta and the probability of that firm being selected as a winner, since β_{di} has a different impact on the two components of probability when $\lambda_t \geq 0$ and when $\lambda_t < 0$. This property of beta is summarized in the following proposition.

Proposition 1: *The relationship between β_{di} and the probability of an asset being selected as a winner is dependent on the sign of the market price of risk, $\bar{\lambda}$.*

- *When $\bar{\lambda} = 0$: The probability of an asset being selected as a winner is positively (negatively) associated with beta for assets with beta greater (smaller) than the cross-sectional average*

beta, $\bar{\beta}_d$.

$$\frac{\partial Pr_i^h}{\partial \beta_{di}} \begin{cases} > 0 & \text{if } \beta_{di} > \bar{\beta}_d \\ = 0 & \text{if } \beta_{di} = \bar{\beta}_d \\ < 0 & \text{if } \beta_{di} < \bar{\beta}_d \end{cases}$$

In addition, suppose that $\beta_{di} > \bar{\beta}_d$ and $\beta_{dj} < \bar{\beta}_d$ where $|\beta_{di} - \bar{\beta}_d| = |\beta_{dj} - \bar{\beta}_d|$: i.e., the distance between β_{di} and $\bar{\beta}$ is identical to the distance between β_{dj} and $\bar{\beta}$. Then $Pr_i^h = Pr_j^h$.

- When $\bar{\lambda} > (<)0$: The probability is positively (negatively) associated with beta for assets with beta greater (smaller) than the cross-sectional average beta. The association can be either positive or negative for other assets. For a sufficiently large, positive (negative) value of market price of risk, $\bar{\lambda}$, the relationship is positive (negative) for an entire range of β_{di} : i.e.,

$$\frac{\partial Pr_i^h}{\partial \beta_{di}} \begin{cases} > 0 (= \text{indeterminate}) & \text{if } \beta_{di} \geq \bar{\beta}_d \\ = \text{indeterminate} (< 0) & \text{if } \beta_{di} < \bar{\beta}_d \end{cases}$$

In addition, suppose that $\beta_{di} > \bar{\beta}_d$ and $\beta_{dj} < \bar{\beta}_d$ where $|\beta_{di} - \bar{\beta}_d| = |\beta_{dj} - \bar{\beta}_d|$. Then $Pr_i^h > Pr_j^h$.

Proof: See Appendix 2.

Corollary 1: When $\bar{\lambda} = 0$, $Pr_{10}^h = Pr_1^l$. If $\bar{\lambda} > (<)0$, $Pr_{10}^h > (<)Pr_1^h$.

Proof: See Appendix 2.

In Figure 3B, we use the results of Lemma 1 to plot the theoretical relation between the predicted probability of security i being chosen a winner, conditional on the beta of security i . The beta in this figure is the asset's beta with respect to the 'true' pricing kernel. Note that if the risk premium is zero, the curve is symmetric around the average value of beta. That is, a positive difference between security i 's beta and the average will increase the probability of being chosen a winner by exactly the same amount as a negative difference of the same absolute value. More importantly, as the risk premium becomes positive and increases, the curve tilts to the right; securities with relatively large betas have a higher probability of being chosen a winner security. A similar interpretation can be made for losers as in Figure 3C.

Thus, the above proposition along with the corollary leads to the result that the functional relationship between β_{di} and Pr_i^h critically depends on the value of the market price of risk $\bar{\lambda}$. If the average market price of risk is zero, the functional relationship is a symmetric “U” shape. In contrast, when $\bar{\lambda}$ is positive, the shape can be tilted to the right, and may even be a monotonically increasing function given a sufficiently large value of $\bar{\lambda}$. That is, assuming stationary risk measures and a positive average market-wide risk premium, the beta of a winner portfolio should, on average, be higher than the beta of a loser portfolio, provided that the beta risk is measured with respect to the proper pricing kernel. Consequently, we expect that momentum strategies will have positive risk.

The tilt in the curve as the risk premium increases simply reflects the increase in security i 's expected return as beta increases. Thus, on average, with a positive market-wide risk premium, the beta of a portfolio formed from past winners should be relatively high, since a portion of that portfolio's high relative return is due to high expected return. This is an extension of the analysis of Lo and MacKinlay (1990), who show that momentum (contrarian) strategies have, even in the absence of time-series effects, an average return which is positive (negative). Note that this result does not imply that abnormal returns cannot be earned with a momentum strategy—merely that the abnormal returns must be earned after allowing for the (on average) positive risk such a strategy entails.

Figure 3D illustrates the theoretical relationship between the risk premium and the expected returns on past winners and past losers. When $\bar{\lambda} = 0$, both types of securities have zero expected return. As the market price of risk increases, the winner's expected return displays an increasing function with a convex shape, while the loser's expected return represents an increasing function with a concave shape. Thus as the market price of risk increases, the expected returns on the two extreme performers diverge, due to the expected difference in their risk measures.

We use this posited relationship between betas and the probability of being chosen as winner or loser as a diagnostic for the “admissibility” of the pricing kernels we consider. Specifically, we analyze the relationship between the estimated probabilities that individual firms are winners and losers and their betas with respect to both the estimated pricing kernels from Section 4 and the market portfolio. The probability is measured as the ratio of the number of periods that stock i

is classified as a winner to the total number of periods considered.

Results of this estimation are summarized in Table 9 and Figure 4. As predicted by Lemma 1, we find a positive relationship between firms' kernel betas and the probability that the firm is a winner for all four estimation techniques used (LOP or NA, unconditional or conditional). These relations are shown in the left-most columns of Table 9 and the top 4 panels of the Figure. The differences between the loser and the winner portfolio betas are striking: for the conditional LOP measure, the winner portfolio beta is more than twice the loser portfolio beta. The difference using the other estimation techniques is also substantial. These results also suggest that the incorporation of conditioning information in the performance measure is consistent with a risk-based explanation for the strategy profits. The incorporation of conditioning variables magnifies the spread in betas (and hence the risk premium) with respect to the stochastic discount factor, and implies that the returns to the momentum strategies examined represent compensation for bearing increased risk.

In contrast, firms' market betas are positively related to *both* the probability that a firm is a winner and a loser. Thus, using this diagnostic, the CAPM kernel appears to be misspecified—the market betas of loser firms are 'too high'. In fact, note that the market betas of losers are slightly larger than the market betas of winners in this sample, consistent with the results of Jegadeesh and Titman (1993). This results in the market beta of the combined (winner minus loser) portfolio being slightly negative, which would correspond to a negative market price of risk, $\bar{\lambda}$, and magnify the risk-adjusted profit of the strategy. Since the market price of risk implied by the CAPM is positive, this result seems to contradict the theory. In addition, this result contributes to the inference that the risk-adjusted return of the momentum strategy is positive. The difference in results across the two types of pricing kernels in this context is another illustration of the importance of benchmark specification in assessing abnormal profits; as Roll (1978) shows, even small differences in benchmarks can lead to large differences in inferences about performance.

5.2 Industry Momentum and Individual Security Momentum

Moskowitz and Grinblatt (1999) document that industry momentum strategies are profitable. The authors find profits of 43 basis points per month in a strategy that buys an equally weighted portfolio of the top three performing industries and shorts an equally weighted portfolio of the

bottom three performing industries, holding this position for six months. The authors attribute these profits to industry components in the data generating process that are unrelated to priced factors, and present evidence that industry momentum is responsible for much of the momentum in individual securities. The authors suggest that their results are potentially attributable to either behavioral or rational explanations for the profitability of momentum strategies.

In this section, we investigate whether our pricing kernel approach can explain the profits to these industry momentum portfolios. The industry portfolios employed in our study are the same as those investigated in Grinblatt and Moskowitz (1999), with the exception that our industry portfolios are equally-weighted rather than value-weighted. We find that using equally-weighted portfolios and our sample period results in slightly lower returns, although these returns are still statistically significant. For example, the return to the 6-month/6-month industry momentum strategy is 27 basis points per month (t -stat=2.60).

Table 10 displays the results of the *LOP* and *NA* performance evaluation for the six-month/six-month industry momentum strategy. When profits are measured relative to the reference assets, the point estimate of profits remains roughly the same at 26 basis points (unconditional) and 28 basis points (conditional), although they are not statistically significant at the 5% or 10% level. These results suggest that while our basis assets, particularly in a conditional setting, can explain individual security momentum profits, they cannot explain the profits of the industry momentum strategy. This appears to confirm the contention of Grundy and Martin (1999) who argue that individual and industry momentum are distinct phenomena. Moreover, the evidence that passive industry (or industry plus managed) portfolios cannot explain “industry momentum” suggests that it is the dynamic nature of the strategy, rather than the industry component, that is important.¹⁵

In a further robustness check, we investigate whether the inclusion of winner/loser securities in the industry portfolios contribute to our finding that momentum portfolio profits are explained by a set of industry portfolios. Specifically, we identify individual winner/loser securities and remove them from their respective industry portfolios. We repeat the analysis of momentum strategy abnormal performance; the results of joint tests are shown in Table 11. Clearly, removing

¹⁵ When industry portfolios are reformed every month, instead of annually, we find that industry momentum profits can be explained by basis assets. Again, this suggests that it is the dynamic nature of the strategy, rather than the industry component, that is important.

individual securities with extreme price movements from the basis does not change our results—we find very little evidence of residual profits in the conditional setting.

The results of this section suggest that the industry momentum strategy of Moskowitz and Grinblatt (1999) cannot be explained by a set of basis assets that consists of a passive weighting of either industry, or industry plus managed, portfolios. That is, the pricing kernel that prices returns to industry portfolios cannot also price the returns to the industry momentum strategy in a dynamic setting. In addition, our evidence on the relation between individual security momentum profits and industry portfolios does not appear to be driven by the inclusion of winner/loser securities in the industry portfolios themselves.

5.3 Monte Carlo Results

In this section, we investigate the power and size characteristics of our tests of abnormal performance in a finite sample, in both the unconditional and conditional settings. We perform a series of Monte Carlo experiments; the details of our data generation are presented in Appendix 3. Briefly, we first generate a time-series of normally distributed returns, using parameters estimated from the observed data, and matching the length of the time-series available in the real data (395 months). We extract a pricing kernel from these data, add a normally distributed shock to the extracted kernel and use the result to form 500 samples of simulated returns. In the conditional setting, we follow a similar procedure, but begin by modelling the conditioning variables and then letting returns depend on these instruments in a manner that is consistent with the observed dependencies in the data. We examine power by constructing two alternative hypotheses: the first is designed to have abnormal performance of 0.5% per month and the second includes an abnormal performances of 1% per month; the larger abnormal performance measure is consistent with the abnormal profits documented in Jegadeesh and Titman (1993).

The results for 5% and 10% tests are presented in Table 12. There are some size distortions, with a rejection rate of 6.8% (12.4%) for the 5% (10%) tests in the unconditional setting, but these distortions are small. The size results in the conditional setting are very similar. The power is relatively high—for the unconditional tests at the smaller abnormal performance level of 50 basis points per month, the power at a significance level of 5% (10%) is 37.8% (48%) for the LOP measures, and 38% (51%) for the NA measure. Not surprisingly, in the conditional setting

with its additional moment conditions, the power declines, but the decline is not large—at the 5% (10%) level for the LOP measure, the power is 30.8% (42.8%), while for the 5% (10%) levels and the NA measure, the power is 30% (41.6%).) When the abnormal performance is 1% per month, the power improves markedly. At the 5% level, the unconditional measures have a power of 86.8% and 83.6%, respectively, for the LOP and NA measures; the conditional tests again experience a slight decline in power, to levels of 77.6% and 80.6% for the LOP and NA measures.

Dahlquist and Soderlind (1999) also conduct Monte Carlo simulations of these test statistic in a finite sample, employing five portfolios and 3 instrumental variables. They find that, depending on the size of their sample and excess return, the power varies between 20% and 70%. The power of the test statistic in our sample, with an abnormal return of 1%, is better than that of the tests they examine; for an abnormal performance of 0.5% per month, they are comparable. Particularly given the larger number of portfolios (and hence moment conditions) that we employ, these results are encouraging. The small sample properties of our test statistic also appear to be superior to those of the test statistics investigated in Bekaert and Urias (1996).

6 Discussion and Interpretation of Results

In the tests considered thus far, we ask whether a pricing kernel exists that (1) satisfies the law of one price, or the more restrictive no-arbitrage condition and (2) prices a momentum portfolio. The answer appears to be ‘yes’: in an unconditional, fixed-weight setting, the abnormal performance of the ‘average’ momentum strategy declines by approximately 48%, is significant in fewer individual cases and is insignificant overall, while in a conditional setting, the abnormal performance is not significantly different from zero in any of the strategies we examine. Using the CAPM benchmark, as Jegadeesh and Titman (1993) do, however, the momentum strategy appears to earn abnormal returns in virtually every strategy they examine. Given the difference in results, it may be helpful to compare the two approaches further. Fortunately, as DeSantis (1995) and Bekaert and Urias (1996) discuss in detail, there is a direct relation between the LOP measure of abnormal performance we use here and the more familiar mean-variance analysis.

In the CAPM, the market is mean-variance efficient, that is, it is the tangency portfolio with the highest possible Sharpe ratio. In contrast, in the LOP estimation procedure that we use, the

20 industry portfolios are used to form an ‘efficient frontier’, relative to which the market portfolio may or may not compare favorably. The test of abnormal performance in this context is a test of whether the momentum portfolio’s returns represent a significant improvement over the opportunity set formed by the industry portfolios. Figure 5 presents the opportunity set created by the industry portfolios, the six-month/six-month momentum strategy and (for comparison) the value-weighted market.

It is clear from this Figure that the market portfolio is dominated by combinations of the industry portfolios that we use. For example, the Sharpe ratio of the market portfolio is 0.1096; in contrast, the maximum Sharpe ratio estimated from the set of industry portfolios is substantially higher. The maximum Sharpe ratio constructed from the industry portfolios is 0.3255; corrected for the bias caused by searching for the maximum (as outlined in MacKinlay (1995)), it is 0.1961. In comparison, the 6-month/6-month momentum portfolio has a Sharpe measure of 0.2084. Thus, the fact that the abnormal performance of momentum portfolios is significant with respect to the CAPM pricing kernel, but declines substantially when industry-sorted portfolios are used to form the benchmark, is a consequence of the fact that the opportunity set created by industry portfolios dominates the market portfolio.

The finding that the Sharpe ratio obtained from the sample of industry portfolios is larger than that observed using the market proxy is consistent with the evidence in Cohen and Polk (1998). MacKinlay (1995) argues that ratios larger than 0.175 are implausibly high; the ratio observed from the industry-sorted portfolios in our sample exceeds that threshold. Thus, while our evidence suggests that momentum portfolio strategies are not mispriced relative to industry portfolios, it may be that industry portfolios are themselves mispriced. In that case, we have succeeded only in shifting the puzzle to another venue. Since the use of industry portfolios is widespread both among academic and practitioners, however, this puzzle is an important one.

If we accept the industry-sorted portfolios as a valid set of basis assets, note that the higher Sharpe measure they imply suggests that it would be feasible to construct a linear multifactor model which prices momentum strategies. For example (and independent of the literature on momentum trading strategies), Dittmar (1999) develops a pricing model based on investors’ preferences in which kurtosis measures are associated with a positive risk premium. Using this *particular* parametric

pricing model as the benchmark and industry portfolios as the set of benchmark assets, we cannot reject the hypothesis that momentum portfolios earn zero abnormal return ($\alpha = 0.0023$, $p\text{-value} = 0.23$).

7 Conclusions and Extensions

This paper investigates the profitability of momentum trading strategies by using a nonparametric test that asks whether these profits can be explained by an equilibrium pricing model that satisfies some minimal restrictions. We investigate two benchmark cases for such a pricing model: the first requires only that the law of one price hold and the second that a no-arbitrage condition hold. We find that a stochastic discount factor can be constructed from a basis set of industry-sorted portfolios which does a relatively good job of explaining momentum profits in an unconditional setting. That is, a risk measure which assumes only that the law of one price holds can account for an average of 48% of the profitability of the trading strategies when agents cannot use conditioning information in forming their expectations. A joint test on the significance of the remaining profits fails to reject the null hypothesis of no abnormal performance. If we require that the stochastic discount factor be positive, the results are similar; some individual momentum strategies retain residual profitability, but the average profits are significantly lower and a joint test of significance fails to reject.

In further tests, when agents are allowed to implement dynamic trading strategies based only on a limited set of public information, we find that the abnormal performance of virtually all of the strategies we consider can be explained. These results suggest the possibility that the returns to momentum strategies are due to the risk of the strategies, rather than investor underreaction. Moreover, the improvement in the results when (a limited set of) conditioning information is allowed to affect investors' expectations suggests that some proportion of momentum "profits" may reflect time-varying risk premia.

It is important to note that, although we avoid assuming that a particular parametric pricing model hold, we are requiring that securities markets be in equilibrium, i.e., by assuming that the law of one price (or no-arbitrage) hold, we are ruling out mispricing in the basis assets. Similarly, in the conditional tests, we are assuming that the ability (if any) of public information variables to

predict the basis assets' returns cannot be due to a (market-wide) lagged price adjustment. If these assumptions do not hold, then measuring profits in relation to these basis assets, and measuring conditional expected returns with respect to these public variables, results in a mis-specified measure of abnormal performance.¹⁶ However, this criticism would apply to any risk-adjusted measure of performance, since such a measure would have to assume an (equilibrium) pricing model. Finally, we develop a diagnostic measure of the risk adjustment used in momentum strategies, and show that the stochastic discount factor we estimate appears to be a better fit than the pricing models previously used to adjust for the risk of such strategies. Our results provide some evidence that momentum strategies are the result of rational, rather than irrational, pricing behavior.

¹⁶In this case, it may still be useful to know that the abnormal return of the trading strategy reflects mispricing of the (passively managed) basis assets, rather than the dynamic nature of the strategy itself.

Appendix 1

Let F_ϵ , F_λ , and $F_{-\lambda}$ represent the respective cumulative distributions of ϵ_{it} , λ_t and $-\lambda_t$. In addition, let $G_\epsilon = 1 - F_\epsilon$ be the reverse cumulant, and $\phi(\cdot)$ and $\Phi(\cdot)$ represent the standard normal probability density function and distribution function respectively. The approximated probability can be represented as:

$$\begin{aligned}
 Pr_i^h &\approx \text{Probability } [\epsilon_{it} > \eta_i(\text{sign}(\lambda_t))\lambda_t] \\
 &= \int_0^\infty G_\epsilon[(\beta_{d10} - \beta_{di})u]dF_\lambda(u) + \int_0^\infty G_\epsilon[(\beta_{di} - \beta_{d1})v]dF_{-\lambda}(v) \\
 &= \frac{1}{\sigma_\lambda} \int_0^\infty \left[\Phi\left(\frac{-(\beta_{d10} - \beta_{di})u}{\sigma_\epsilon}\right) \phi\left(\frac{u - \bar{\lambda}}{\sigma_\lambda}\right) + \Phi\left(\frac{-(\beta_{di} - \beta_{d1})u}{\sigma_\epsilon}\right) \phi\left(\frac{u + \bar{\lambda}}{\sigma_\lambda}\right) \right] du, \quad (11)
 \end{aligned}$$

Proof: From the independence of ϵ_{it} and λ_t , the first term can be written as:

$$\begin{aligned}
 \text{Prob } [\epsilon_{it} > (\beta_{d10} - \beta_{di})\lambda_t \text{ and } \lambda_t \geq 0] &= \int_0^\infty \int_{(\beta_{d10} - \beta_{di})u}^\infty f_{\epsilon, \lambda}(z, u) dz du \\
 &= \int_0^\infty \int_{(\beta_{d10} - \beta_{di})u}^\infty dF_\epsilon(z) dF_\lambda(u) \\
 &= \int_0^\infty G_\epsilon[(\beta_{d10} - \beta_{di})u] dF_\lambda(u) \\
 &= \frac{1}{\sigma_\lambda} \int_0^\infty \left[1 - \Phi\left(\frac{(\beta_{d10} - \beta_{di})u}{\sigma_\epsilon}\right) \right] \phi\left(\frac{u - \bar{\lambda}}{\sigma_\lambda}\right) du \\
 &= \frac{1}{\sigma_\lambda} \int_0^\infty \Phi\left(\frac{-(\beta_{d10} - \beta_{di})u}{\sigma_\epsilon}\right) \phi\left(\frac{u - \bar{\lambda}}{\sigma_\lambda}\right) du,
 \end{aligned}$$

where $f_{\epsilon, \lambda}(\cdot, \cdot)$ is the joint probability distribution function of ϵ_{it} and λ_t .

Similarly,

$$\begin{aligned}
 \text{Prob } [\epsilon_{it} > (\beta_{d1} - \beta_{di})\lambda_t \text{ and } \lambda_t < 0] &= \int_{-\infty}^0 \int_{(\beta_{d1} - \beta_{di})u}^\infty f_{\epsilon, \lambda}(z, u) dz du \\
 &= \int_0^\infty \int_{(\beta_{d1} - \beta_{di})(-u)}^\infty dF_\epsilon(z) dF_\lambda(-u) \\
 &= \int_0^\infty G_\epsilon[(\beta_{di} - \beta_{d1})v] dF_{-\lambda}(v) \\
 &= \frac{1}{\sigma_\lambda} \int_0^\infty \Phi\left(\frac{-(\beta_{di} - \beta_{d1})u}{\sigma_\epsilon}\right) \phi\left(\frac{u + \bar{\lambda}}{\sigma_\lambda}\right) du.
 \end{aligned}$$

This completes the proof. *q.e.d.* □

Note that the first term in the probability of being characterized as a winner is the probability that the asset with β_{di} is a winner when the market moves in a positive direction (i.e., $\lambda_t > 0$), and

the second term represents the probability that the same asset is a winner when the market moves in the reverse direction (i.e., $\lambda_t < 0$).

In Figure 3-A, we simulate security returns for a particular set of parameters and calculate the probability of being chosen a winner. The Figure suggests that the performance of this approximation is relatively good.

Appendix 2

Proof of Lemma 1: First, the change in the probability of being a ‘winner’ in a positive market with respect to a change in the beta is

$$\begin{aligned}
\frac{\partial \text{Prob}(R_{it} \in H | \lambda_t \geq 0)}{\partial \beta_{di}} &= \left[\frac{\int_0^\infty G_\epsilon[(\beta_{d10} - \beta_{di})u] dF_\lambda(u)}{1 - \Phi(-\frac{\bar{\lambda}}{\sigma_\lambda})} \right] / \partial \beta_{di} \\
&= \frac{1}{1 - \Phi(-\frac{\bar{\lambda}}{\sigma_\lambda})} \int_0^\infty \frac{\partial G_\epsilon[(\beta_{d10} - \beta_{di})u]}{\partial \beta_{di}} dF_\lambda(u) \\
&= \frac{1}{\Phi(\frac{\bar{\lambda}}{\sigma_\lambda})} \int_0^\infty G'_\epsilon[(\beta_{d10} - \beta_{di})u] (-u) dF_\lambda(u) \\
&= \frac{1}{\Phi(\frac{\bar{\lambda}}{\sigma_\lambda})} \int_0^\infty f_\epsilon[(\beta_{d10} - \beta_{di})u] u dF_\lambda(u) \\
&> 0
\end{aligned}$$

where

$$f_\epsilon(x) = \frac{\phi\left(\frac{x}{\sigma_\epsilon}\right)}{\sigma_\epsilon}.$$

The last inequality comes from the fact that the feasible range of u is nonnegative. A similar calculation yields

$$\begin{aligned}
\frac{\partial \text{Prob}(r_{it} \in H | \lambda_t < 0)}{\partial \beta_{di}} &= -\frac{1}{\Phi(-\frac{\bar{\lambda}}{\sigma_\lambda})} \int_0^\infty f_\epsilon[(\beta_{di} - \beta_{d1})v] v dF_{-\lambda}(v) \\
&< 0 \quad q.e.d.
\end{aligned}$$

□

Proof of Proposition 1: From (11),

$$\frac{\partial P_{it}^h}{\partial \beta_{di}} = \int_0^\infty f_\epsilon[(\beta_{d10} - \beta_{di})u] u dF_\lambda(u) - \int_0^\infty f_\epsilon[(\beta_{di} - \beta_{d1})v] v dF_{-\lambda}(v). \quad (12)$$

Beginning with the first term in (12),

$$\begin{aligned}
\int_0^\infty f_\epsilon[(\beta_{d10} - \beta_{di})u] u dF_\lambda(u) &= \frac{1}{\sigma_\lambda \sigma_\epsilon} \int_0^\infty \phi\left(\frac{(\beta_{di} - \beta_{d10})u}{\sigma_\epsilon}\right) \phi\left(\frac{u - \bar{\lambda}}{\sigma_\lambda}\right) u du \\
&= \frac{\sigma_{\psi_1}}{\sigma_\epsilon \sigma_\lambda} \phi\left(\frac{(\beta_{d10} - \beta_{di})\bar{\psi}_1 \sigma_\lambda}{\sigma_\epsilon \sigma_{\psi_1}}\right) \Phi\left(\frac{\bar{\psi}_1}{\sigma_{\psi_1}}\right) (\bar{\psi}_1 + 2\sigma_{\psi_1} \phi(0)),
\end{aligned}$$

$$\text{where } \bar{\psi}_1 = \frac{\sigma_\epsilon^2 \bar{\lambda}}{(\beta_{d10} - \beta_{di})^2 \sigma_\lambda^2 + \sigma_\epsilon^2}, \quad \sigma_{\psi_1}^2 = \frac{\sigma_\epsilon^2 \sigma_\lambda^2}{(\beta_{d10} - \beta_{di})^2 \sigma_\lambda^2 + \sigma_\epsilon^2}.$$

A similar computation yields the second term:

$$\int_0^\infty f_\epsilon[(\beta_{di} - \beta_{d1})v]v dF_{-\lambda}(v) = \frac{\sigma_{\psi_2}}{\sigma_\epsilon \sigma_\lambda} \phi\left(\frac{(\beta_{di} - \beta_{d1})\bar{\psi}_2 \sigma_\lambda}{\sigma_\epsilon \sigma_{\psi_2}}\right) \Phi\left(\frac{\bar{\psi}_2}{\sigma_{\psi_2}}\right) (\bar{\psi}_2 + 2\sigma_{\psi_2} \phi(0)),$$

$$\text{where } \bar{\psi}_2 = -\frac{\sigma_\epsilon^2 \bar{\lambda}}{(\beta_{di} - \beta_{d1})^2 \sigma_\lambda^2 + \sigma_\epsilon^2}, \quad \sigma_{\psi_2}^2 = \frac{\sigma_\epsilon^2 \sigma_\lambda^2}{(\beta_{di} - \beta_{d1})^2 \sigma_\lambda^2 + \sigma_\epsilon^2}.$$

- When $\bar{\lambda} = 0$: $\bar{\psi}_1 = \bar{\psi}_2 = 0$, and thus

$$\frac{\partial Pr_i^h}{\partial \beta_{di}} = (\sigma_{\psi_1}^2 - \sigma_{\psi_2}^2) \frac{\phi(0)^2}{\sigma_\epsilon \sigma_\lambda}.$$

Therefore, $\text{sign}\left(\frac{\partial Pr_i^h}{\partial \beta_{di}}\right) = \text{sign}\left(\sigma_{\psi_1}^2 - \sigma_{\psi_2}^2\right)$. From the definition of $\sigma_{\psi_1}^2$ and $\sigma_{\psi_2}^2$, it can be shown that

$$\sigma_{\epsilon_1}^2 \geq (<) \sigma_{\epsilon_1}^2 \quad \text{iff } (\beta_{d10} - \beta_{di})^2 - (\beta_{di} - \beta_{d1})^2 \leq (>) 0.$$

Note that $(\beta_{d10} - \beta_{di})^2 - (\beta_{di} - \beta_{d1})^2$ measures the difference in distances of β_{di} to β_{d10} and β_{d1} . As such, $(\beta_{d10} - \beta_{di})^2 - (\beta_{di} - \beta_{d1})^2 \leq (>) 0$ if $\beta_{di} \geq (<) \bar{\beta}_d$. Further, the magnitude of $|\frac{\partial Pr_i^h}{\partial \beta_{di}}|$ is determined by the distance of β_{di} to $\bar{\beta}_d$, Pr_i^h as a functional of β_{di} displays a symmetric ‘U’ shape.

- When $\bar{\lambda} > 0$: Consider $\beta_{di} = \bar{\beta}_d$. In this case $\bar{\psi}_1 = -\bar{\psi}_2 > 0$, and $\sigma_{\psi_1}^2 = \sigma_{\psi_2}^2$, which yields

$$\frac{\partial Pr_i^h}{\partial \beta_{di}} = \frac{\sigma_{\psi_1}}{\sigma_\epsilon \sigma_\lambda} \Phi\left[(\beta_1 - \bar{\beta})\bar{\psi}_1 \frac{\sigma_\lambda}{\sigma_\epsilon \sigma_{\psi_1}}\right] \left[(\bar{\psi}_1 + 2\sigma_{\psi_1} \phi(0)) \Phi\left(\frac{\bar{\psi}_1}{\sigma_{\psi_1}}\right) - (-\bar{\psi}_1 + 2\sigma_{\psi_1} \phi(0)) \Phi\left(-\frac{\bar{\psi}_1}{\sigma_{\psi_1}}\right) \right].$$

Note that $\bar{\psi}_1 > -\bar{\psi}_1$ and $\Phi\left(\frac{\bar{\psi}_1}{\sigma_{\psi_1}}\right) > \Phi\left(-\frac{\bar{\psi}_1}{\sigma_{\psi_1}}\right)$. Thus $\frac{\partial Pr_i^h}{\partial \beta_{di}} > 0$ even for the asset with the average beta asset. Therefore, the shape of Pr_i^h as a functional of β_{di} tilts to the right. q.e.d

□

Proof of Corollary 1: From (11), it can be shown that

$$Pr_{10}^h = \frac{1}{\sigma_\lambda} \left[\frac{1}{2} \int_0^\infty \phi\left(\frac{u - \bar{\lambda}}{\sigma_\lambda}\right) du + \int_0^\infty \Phi\left(-\frac{(\beta_{d10} - \beta_{d1})u}{\sigma_\epsilon}\right) \phi\left(\frac{u + \bar{\lambda}}{\sigma_\lambda}\right) du \right] \quad (13)$$

$$Pr_1^h = \frac{1}{\sigma_\lambda} \left[\frac{1}{2} \int_0^\infty \phi\left(\frac{u + \bar{\lambda}}{\sigma_\lambda}\right) du + \int_0^\infty \Phi\left(-\frac{(\beta_{d10} - \beta_{d1})u}{\sigma_\epsilon}\right) \phi\left(\frac{u - \bar{\lambda}}{\sigma_\lambda}\right) du \right] \quad (14)$$

- When $\bar{\lambda} = 0$: Substituting $\bar{\lambda} = 0$ into (13) and (14) leads to $Pr_{10}^h = Pr_1^h$.

- When $\bar{\lambda} > (<)0$: Given that $Pr_{10}^h = Pr_1^h$ when $\bar{\lambda} = 0$, we should show $\frac{\partial(Pr_{10}^h - Pr_1^h)}{\partial \bar{\lambda}} > (<)0$ when $\bar{\lambda} > (<)0$.

$$\begin{aligned}
\frac{\partial (Pr_{10}^h - Pr_1^h)}{\partial \bar{\lambda}} &= \frac{1}{\sigma_\lambda} \frac{1}{2} \int_0^\infty \left[\frac{\bar{\lambda}}{\sigma_\lambda} \phi \left(\frac{u - \bar{\lambda}}{\sigma_\lambda} \right)^2 + \frac{\lambda}{\sigma_\lambda} \phi \left(\frac{u + \bar{\lambda}}{\sigma_\lambda} \right)^2 \right] du \\
&\quad + \frac{1}{\sigma_\lambda} \int_0^\infty \Phi \left(-\frac{(\beta_{d10} - \beta_{d1})u}{\sigma_\epsilon} \right) \left[-\frac{\bar{\lambda}}{\sigma_\lambda} \phi \left(\frac{u + \bar{\lambda}}{\sigma_\lambda} \right)^2 - \frac{\bar{\lambda}}{\sigma_\lambda} \phi \left(\frac{u - \bar{\lambda}}{\sigma_\lambda} \right)^2 \right] du \\
&= \int_0^\infty \bar{\lambda} \left[\frac{1}{2} - \Phi \left(-\frac{(\beta_{d10} - \beta_{d1})u}{\sigma_\epsilon} \right) \right] \left[\phi \left(\frac{u + \bar{\lambda}}{\sigma_\lambda} \right)^2 + \phi \left(\frac{u - \bar{\lambda}}{\sigma_\lambda} \right)^2 \right] du
\end{aligned}$$

Since $\Phi \left(-\frac{(\beta_{d10} - \beta_{d1})u}{\sigma_\epsilon} \right) < \frac{1}{2} = \Phi(0)$, all the arguments are positive except for $\bar{\lambda}$ in the above integral. Therefore,

$$sign \left(\frac{\partial (Pr_{10}^h - Pr_1^h)}{\partial \bar{\lambda}} \right) = sign(\bar{\lambda}).$$

This ends the proof. *q.e.d.* □

Appendix 3 - Monte Carlo Experiments

Unconditional Measure: The unconditional return series is generated by assuming $R \sim N(\mu, \Sigma)$, where μ represents the mean and Σ represents the covariance matrix of the observed returns data. We generate 395 time series and 21 cross-sectional observations to match the data used in the paper. The vector of returns is then generated as

$$\mathbf{R} = \mu + \mathbf{c}\sqrt{\Lambda}\mathbf{c}'\mathbf{u}$$

where c represents the characteristic (normalized) eigenvector of Σ , Λ represents the eigenvalue matrix of Σ , and u is a conforming vector of standard normal variates. The return vector is then used to create an economy-wide pricing kernel, $m = R'E[RR']^{-1}1_N$. We then add a normally distributed shock to this pricing kernel, and use the result to form test asset returns. Under the null of no abnormal pricing, the return on the test assets can be described as

$$r = \frac{Cov[m, r]}{E[m]} + \epsilon$$

where $\epsilon \sim N(0, \sigma(r))$. We choose $Cov[m, r]$ and $\sigma(r)$ to match the standard deviation of the 6-month/6-month strategy and the covariance of the strategy with the empirical pricing kernel. We construct our alternative as

$$r = 0.01 + \frac{Cov[m, r]}{E[m]} + \epsilon$$

Thus, under the alternative, the strategy displays a 100 basis point monthly abnormal performance.

Conditional Measure: The conditional measure experiments are formed similarly to the unconditional measure. However, we start the procedure by modeling the conditioning variables

$$\mathbf{Z} = \mu(\mathbf{Z}) + \mathbf{c}(\mathbf{Z})\sqrt{\Lambda(\mathbf{Z})}\mathbf{c}(\mathbf{Z})'\mathbf{u}$$

where $\mu(Z)$ represents the mean of the observed conditioning variables, $c(Z)$ and $\Lambda(Z)$ represent the normalized characteristic vector and eigenvalue matrix of the covariance matrix of the instruments, respectively, and u is a conforming vector of standard normal variates. The returns are generated as

$$\mathbf{R} = \mathbf{Z}\delta + \mathbf{c}(\mathbf{R})\sqrt{\Lambda(\mathbf{R})}\mathbf{c}(\mathbf{R})'\mathbf{v}$$

where δ represents the coefficients of a projection of the observed returns on the observed instruments, $R = Z\delta + \epsilon$, $\epsilon \sim N(0, \Omega)$. The terms $c(R)$ and $\Lambda(R)$ represent the normalized characteristic

vector and eigenvalue matrix of the observed residual covariance matrix, Ω . The strategy returns under the null and alternative are created as in the unconditional experiment above.

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Mean Returns				
	K=3	K=6	K=9	K=12
J=3: Win	0.0132	0.0131	0.0148	0.0154
Lose	0.0121	0.0125	0.0099	0.0093
Win-Lose	0.0011	0.0006	0.0049	0.0061
J=6: Win	0.0158	0.0167	0.0172	0.0164
Lose	0.0102	0.0090	0.0081	0.0090
Win-Lose	0.0056	0.0077	0.0091	0.0074
J=9: Win	0.0180	0.0185	0.0175	0.0163
Lose	0.0090	0.0075	0.0081	0.0094
Win-Lose	0.0090	0.0090	0.0094	0.0079
J=12: Win	0.0188	0.0178	0.0168	0.0156
Lose	0.0072	0.0075	0.0087	0.0100
Win-Lose	0.0116	0.0103	0.0081	0.0056

Standard Deviations				
	K=3	K=6	K=9	K=12
J=3: Win	0.0629	0.0628	0.0635	0.0645
Lose	0.0860	0.0850	0.0850	0.0802
Win-Lose	0.0520	0.0484	0.0452	0.0368
J=6: Win	0.0637	0.0644	0.0647	0.0644
Lose	0.0910	0.0883	0.0857	0.0846
Win-Lose	0.0625	0.0567	0.0507	0.0484
J=9: Win	0.0651	0.0651	0.0651	0.0649
Lose	0.0924	0.0905	0.0890	0.0881
Win-Lose	0.0648	0.0612	0.0577	0.0552
J=12: Win	0.0662	0.0657	0.0655	0.0653
Lose	0.0914	0.0911	0.0906	0.0899
Win-Lose	0.0641	0.0627	0.0605	0.0583

Table 1 presents monthly means and standard deviations of the returns to 16 relative strength strategies formed as in Jegadeesh and Titman (1993). The portfolios are formed by purchasing the portfolio formed of firms in the tenth decile of returns and selling the portfolio formed of firms in the first decile of returns over the past J months. Portfolios are held for the subsequent K months. The data cover the period 2/28/65 through 12/31/97.

Table 1: Summary Statistics: Momentum Portfolios

Panel A: Mean Returns			
Industry	Mean Return	Industry	Mean Return
Mining	0.0123	Electrical Equipment	0.0155
Food & Beverage	0.0139	Transport Equipment	0.0142
Textile & Apparel	0.0105	Manufacturing	0.0141
Paper Products	0.0127	Railroads	0.0131
Chemical	0.0143	Other Transportation	0.0136
Petroleum	0.0137	Utilities	0.0101
Construction	0.0131	Department Stores	0.0110
Primary Metals	0.0106	Other Retail	0.0133
Fabricated Metals	0.0149	Finance, Real Estate	0.0117
Machinery	0.0129	Other	0.0150

Panel B: Standard Deviations			
Industry	Standard Dev.	Industry	Standard Dev.
Mining	0.0672	Electrical Equipment	0.0736
Food & Beverage	0.0493	Transport Equipment	0.0653
Textile & Apparel	0.0687	Manufacturing	0.0675
Paper Products	0.0583	Railroads	0.0542
Chemical	0.0521	Other Transportation	0.0673
Petroleum	0.0561	Utilities	0.0370
Petroleum	0.0611	Department Stores	0.0680
Construction	0.0586	Other Retail	0.0626
Fabricated Metals	0.0616	Finance, Real Estate	0.0588
Machinery	0.0641	Other	0.0645

Table 2 presents monthly means and standard deviations of the returns to 20 industry-sorted portfolios as in Moskowitz (1996). Portfolios are formed on the basis of 2-digit SIC codes and are equally-weighted, rebalanced annually. The data cover the period 2/28/65 through 12/31/97.

Table 2: Summary Statistics: Industry Portfolios

	J=3, K=3	J=3,K=6	J=3, K=9	J=3,K=12
α_{W-L}	0.0021	0.0015	0.0035	0.0050
p	(0.4306)	(0.5229)	(0.1087)	(0.0073)
	J=6, K=3	J=6,K=6	J=6, K=9	J=6,K=12
α_{W-L}	0.0065	0.0085	0.0082	0.0068
p	(0.0377)	(0.0033)	(0.0015)	(0.0058)
	J=9, K=3	J=9,K=6	J=9, K=9	J=9,K=12
α_{W-L}	0.0099	0.0094	0.0085	0.0063
p	(0.0028)	(0.0026)	(0.0038)	(0.0245)
	J=12, K=3	J=12,K=6	J=12, K=9	J=12,K=12
α_{W-L}	0.0123	0.0103	0.0082	0.0057
p	(0.0002)	(0.0013)	(0.0076)	(0.0528)

Table 3 presents results from estimating the excess returns on the relative strength strategies based on a Jensen's (1968) performance measure. Specifically, the table presents estimates of the intercepts from the least squares regression

$$r_{TS,t+\tau} = \alpha_{TS} + \beta_{TS}(R_{M,t+\tau} - R_{f,t+\tau}) + \epsilon_{TS,t+\tau},$$

where $r_{TS,t+\tau}$ denotes the excess return on relative strength portfolio TS and $R_{M,t+\tau}$ represents the return on the value-weighted CRSP index over the one-month Treasury Bill return.

Table 3: CAPM Performance Measure

	J=3, K=3	J=3,K=6	J=3, K=9	J=3,K=12
α_{W-L}	0.0056	0.0048	0.0082	0.0090
SE	(0.0022)	(0.0021)	(0.0018)	(0.0016)
	J=6, K=3	J=6,K=6	J=6, K=9	J=6,K=12
α_{W-L}	0.0108	0.0123	0.0130	0.0114
SE	(0.0027)	(0.0025)	(0.0022)	(0.0021)
	J=9, K=3	J=9,K=6	J=9, K=9	J=9,K=12
α_{W-L}	0.0144	0.0156	0.0142	0.0119
SE	(0.0028)	(0.0026)	(0.0024)	(0.0022)
	J=12, K=3	J=12,K=6	J=12, K=9	J=12,K=12
α_{W-L}	0.0169	0.0157	0.0137	0.0112
SE	(0.0028)	(0.0027)	(0.0025)	(0.0024)

Table 4 presents results from estimating the excess returns on the relative strength strategies based on a Jensen's (1968) performance measure. Specifically, the table presents estimates of the intercepts from the least squares regression

$$r_{TS,t+\tau} = \alpha_{TS} + \beta_{TS,MRP}(R_{MRP,t+\tau}) + \beta_{TS,SMB}(R_{SMB,t+\tau}) + \beta_{TS,MRP}(R_{HML,t+\tau}) + \epsilon_{TS,t+\tau},$$

where $r_{TS,t+\tau}$ denotes the excess return on relative strength portfolio TS , $R_{MRP,t+\tau}$ represents the return on the value-weighted CRSP index over the one-month Treasury Bill return, R_{SMB} represents the return on a portfolio that is long large-capitalization stocks and short in small-capitalization stocks, and R_{HML} represents the return on a portfolio that is long in high book-to-market stocks and short in low book-to-market stocks.. Heteroskedasticity-consistent standard errors are presented in parentheses.

Table 4: Fama-French Performance Measure

	J=3, K=3	J=3,K=6	J=3, K=9	J=3,K=12
α_{W-L}	0.0001	-0.0004	0.0035	0.0036
p	(0.3355)	(0.3119)	(0.1410)	(0.0306)
	J=6, K=3	J=6,K=6	J=6, K=9	J=6,K=12
α_{W-L}	0.0035	0.0048	0.0053	0.0037
p	(0.5406)	(0.1509)	(0.0289)	(0.0867)
	J=9, K=3	J=9,K=6	J=9, K=9	J=9,K=12
α_{W-L}	0.0052	0.0061	0.0047	0.0028
p	(0.0696)	(0.0179)	(0.0427)	(0.2400)
	J=12, K=3	J=12,K=6	J=12, K=9	J=12,K=12
α_{W-L}	0.0066	0.0051	0.0034	0.0016
p	(0.0248)	(0.0600)	(0.2183)	(0.3723)

Table 5 presents results from the unconditional law of one price estimation of performance measures. The table presents results for sixteen strategies representing different combinations of sorting and holding periods. Portfolios are ranked on the basis of J -month lagged returns and held for K months. α represents average monthly excess performance over the portfolio of basis assets for the strategy. χ^2 represents the test of the hypothesis $H_0 : \alpha = 0$ with p -values in parentheses. The basis assets in this sample consist of 20 industry sorted portfolios.

Table 5: LOP-Based Performance Measures - Unconditional

	J=3, K=3	J=3,K=6	J=3, K=9	J=3,K=12
α_{W-L}	0.0001	-0.0003	0.0035	0.0043
p	(0.3446)	(0.3169)	(0.1402)	(0.0307)
	J=6, K=3	J=6,K=6	J=6, K=9	J=6,K=12
α_{W-L}	0.0032	0.0048	0.0051	0.0036
p	(0.5045)	(0.1695)	(0.0297)	(0.0852)
	J=9, K=3	J=9,K=6	J=9, K=9	J=9,K=12
α_{W-L}	0.0053	0.0059	0.0046	0.0028
p	(0.0765)	(0.0194)	(0.0528)	(0.2824)
	J=12, K=3	J=12,K=6	J=12, K=9	J=12,K=12
α_{W-L}	0.0062	0.0049	0.0032	0.0016
p	(0.0264)	(0.0588)	(0.2142)	(0.3920)

Table 6 presents results from the unconditional no arbitrage estimation of performance measures. The table presents results for sixteen strategies representing different combinations of sorting and holding periods. Portfolios are ranked on the basis of J -month lagged returns and held for K months. α represents average monthly excess performance over the portfolio of basis assets for the strategy. χ^2 represents the test of the hypothesis $H_0 : \alpha = 0$ with p -values in parentheses. The basis assets in this sample consist of 20 industry sorted portfolios.

Table 6: NA-Based Performance Measures - Unconditional

	J=3, K=3	J=3,K=6	J=3, K=9	J=3,K=12
α_{W-L}	-0.0017	-0.0020	0.0015	0.0026
α_W	0.0014	0.0021	0.0037	0.0030
α_L	0.0031	0.0041	0.0022	0.0004
p	(0.0914)	(0.0553)	(0.0634)	(0.0858)
	J=6, K=3	J=6,K=6	J=6, K=9	J=6,K=12
α_{W-L}	0.0015	0.0027	0.0047	0.0031
p	(0.5378)	(0.3986)	(0.1421)	(0.1550)
	J=9, K=3	J=9,K=6	J=9, K=9	J=9,K=12
α_{W-L}	0.0052	0.0044	0.0034	0.0024
α_W	0.0033	0.0044	0.0057	0.0027
α_L	-0.0019	-0.0000	0.0023	0.0003
p	(0.3246)	(0.1634)	(0.1899)	(0.1944)
	J=12, K=3	J=12,K=6	J=12, K=9	J=12,K=12
α_{W-L}	0.0054	0.0040	0.0032	0.0016
α_W	0.0020	0.0045	0.0021	0.0010
α_L	-0.0034	0.0005	-0.0011	-0.0006
p	(0.1446)	(0.1075)	(0.1352)	(0.1904)

Table 7 presents results from the conditional law of one price estimation of performance measures. The table presents results for sixteen strategies representing different combinations of sorting and holding periods. Portfolios are ranked on the basis of J -month lagged returns and held for K months. α represents average monthly excess performance over the portfolio of basis assets for the strategy. χ^2 represents the test of the hypothesis $H_0 : \alpha = 0$ with p -values in parentheses. The basis assets in this sample consist of 20 industry sorted portfolios, augmented by managed portfolios, which consist of the product of the equity portfolios with the instrumental variables $z_t = \{tb, ts, dy\}$, for a total of 80 basis assets.

Table 7: LOP-Based Performance Measures - Conditional

	J=3, K=3	J=3,K=6	J=3, K=9	J=3,K=12
α_{W-L}	-0.0035	-0.0040	0.0007	0.0023
χ^2	1.0773	1.6094	0.1231	1.5340
p	(0.2993)	(0.2046)	(0.7257)	(0.2155)
	J=6, K=3	J=6,K=6	J=6, K=9	J=6,K=12
α_{W-L}	0.0006	0.0026	0.0035	0.0023
χ^2	0.0374	0.8081	2.3466	0.9456
p	(0.8466)	(0.3687)	(0.1256)	(0.3308)
	J=9, K=3	J=9,K=6	J=9, K=9	J=9,K=12
α_{W-L}	0.0029	0.0043	0.0031	0.0015
χ^2	0.9022	2.9074	1.7524	0.3667
p	(0.3422)	(0.0882)	(0.1856)	(0.5448)
	J=12, K=3	J=12,K=6	J=12, K=9	J=12,K=12
α_{W-L}	0.0051	0.0040	0.0021	0.0000
χ^2	3.0416	2.6617	0.6877	0.0003
p	(0.0812)	(0.1028)	(0.4070)	(0.9854)

Table 8 presents results from the conditional no arbitrage estimation of performance measures. The table presents results for sixteen strategies representing different combinations of sorting and holding periods. Portfolios are ranked on the basis of J -month lagged returns and held for K months. α represents average monthly excess performance over the portfolio of basis assets for the strategy. χ^2 represents the test of the hypothesis $H_0 : \alpha = 0$ with p -values in parentheses. The basis assets in this sample consist of 20 industry sorted portfolios, augmented by managed portfolios, which consist of the product of the equity portfolios with the instrumental variables $z_t = \{tb, ts, dy\}$, for a total of 80 basis assets.

Table 8: NA-Based Performance Measures - Conditional

Decile	Mean	LOP-Uncond.	NA-Uncond.	LOP-Cond.	NA-Cond.	Value-Weighted
1	0.0090	1.4411	1.3673	0.5863	1.2504	1.3797
2	0.0013	1.5027	1.4555	0.6634	1.3512	1.1784
3	0.0125	1.4987	1.4615	0.7702	1.4298	1.1095
4	0.0126	1.4532	1.4250	0.8610	1.4011	1.0726
5	0.0127	1.4328	1.4107	0.9399	1.4611	1.0474
6	0.0131	1.4689	1.4547	0.9615	1.4881	1.0417
7	0.0134	1.5184	1.5069	1.0351	1.6054	1.0475
8	0.0137	1.5955	1.5872	1.1032	1.7612	1.0779
9	0.0146	1.7415	1.7316	1.1915	1.9192	1.1369
10	0.0167	1.9379	1.9203	1.3230	2.2963	1.2588

Table 9 presents betas and mean returns for the decile portfolios of the 6-month/6-month momentum strategy. Betas are estimated with respect to the 6-month/6-month unconditional LOP, unconditional NA, conditional LOP, and unconditional NA pricing kernels and the value-weighted portfolio.

Table 9: Portfolio Betas

Panel A: Unconditional

Measure:	<i>LOP</i>	<i>NA</i>
α_{W-L}	0.0026	0.0026
χ^2	1.3233	1.3120
p	(0.2500)	(0.2520)

Panel B: Conditional

Measure:	<i>LOP</i>	<i>NA</i>
α_{W-L}	0.0028	0.0021
χ^2	1.9918	0.8285
p	(0.1581)	(0.3627)

Table 10 presents results from the LOP and NA performance measures in unconditional (Panel A) and conditional (Panel B) settings for the 6-month/6-month *industry* momentum strategy. α represents average monthly excess performance over the portfolio of basis assets for the strategy. χ^2 represents the test of the hypothesis $H_0 : \alpha = 0$ with p -values in parentheses. The basis assets in this sample consist of 20 industry sorted portfolios (for unconditional measures), augmented by managed portfolios (for conditional measures), which consist of the product of the equity portfolios with the instrumental variables $z_t = \{tb, ts, dy\}$, for a total of 80 basis assets.

Table 10: Performance Assessment of Industry Momentum Strategies

	LOP-Uncond.	NA-Uncond.	LOP-Cond.	NA-Cond.
p	(0.453)	(0.440)	(0.530)	(0.371)

Table 11 presents results of joint tests for the significance of 16 momentum strategies using both the LOP and NA measures in unconditional and conditional settings when winners and losers are removed from basis assets. The p -values shown are the result of testing the hypothesis $H_0 : \alpha = 0$. The basis assets in this sample consist of 20 industry sorted portfolios (for unconditional measures), augmented by managed portfolios (for conditional measures), which consist of the product of the equity portfolios with the instrumental variables $z_t = \{tb, ts, dy\}$, for a total of 80 basis assets.

Table 11: Performance Assessment: Winners and Losers Removed from Basis

Measure	50 bp		100bp
	10(5)% Size	10(5)% Power	10(5)% Power
LOP-U	12.4(6.8)	48.0(37.8)	92.6(86.8)
NA-U	15.4(8.8)	51.0(38.0)	93.8(83.6)
LOP-C	12.8(6.0)	42.8(30.8)	85.2(77.6)
NA-C	13.2(6.0)	41.6(30.0)	87.6(80.6)

Table 12 presents the empirical size and power of the test statistics used in the paper. Column 1 presents the size results at 5% and 10% levels. The n% size test represents the percent of tests that reject the null under the hypothesis of no abnormal performance at the n% significance level. Columns 2 and 3 present the power of the test statistic at different levels of abnormal performance. Column 2 is the percentage of rejections at 5% and 10% levels, with abnormal performance of 50 basis points per month in the simulated data. Column 3 is the percentage of rejections at 5% and 10% levels, with abnormal performance of 100 basis points per month in the simulated data. 500 Monte Carlo replications are employed throughout. Details of the Monte Carlo experiments are presented in Appendix 3.

Table 12: Monte Carlo Analysis of Performance Measure Distributions

Figure 1: Unconditional Hansen-Jagannathan Bounds: Industry and Momentum Portfolios

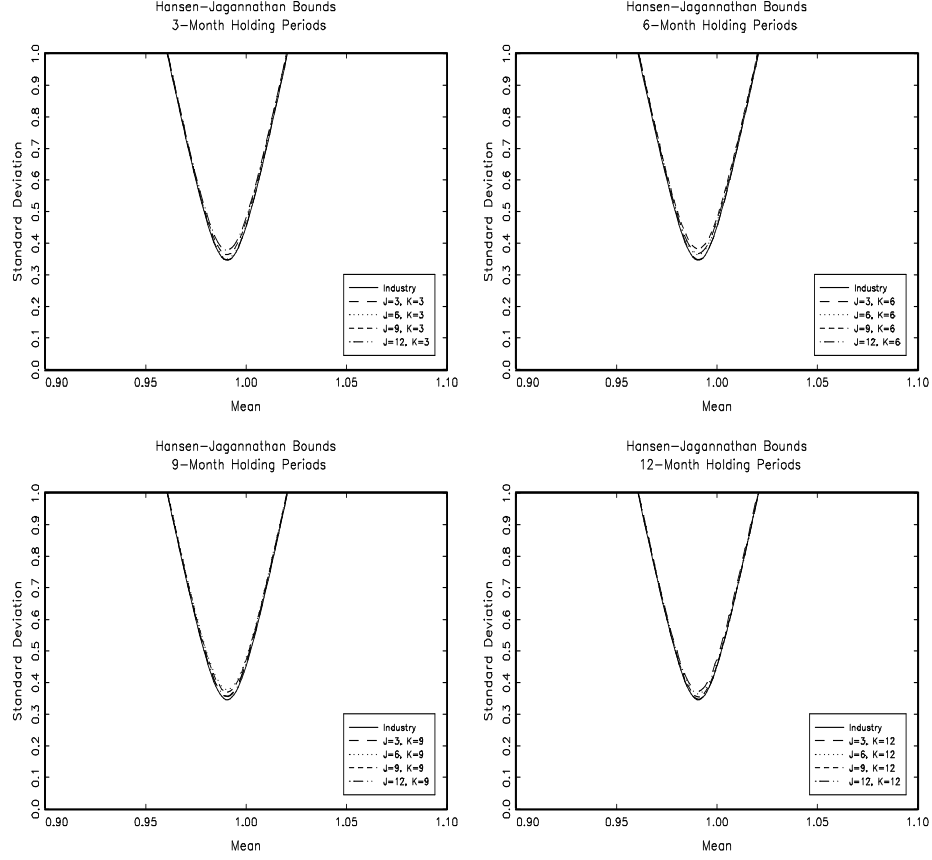


Figure 1 displays the unconditional Hansen and Jagannathan (1991) bounds on the mean and standard deviation of admissible pricing kernels for the 20 industry-sorted portfolios and the momentum portfolios. Momentum portfolios are grouped by their holding periods $K = 3, 6, 9, 12$ months.

Figure 2: Conditional Hansen-Jagannathan Bounds: Industry and Momentum Portfolios

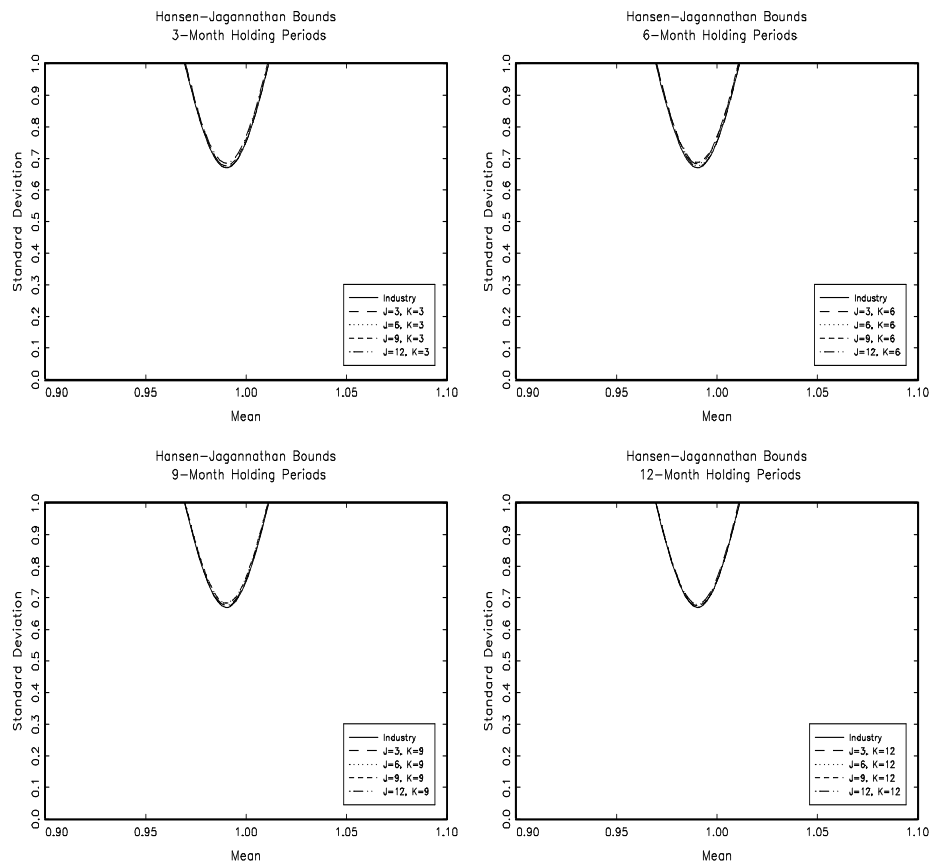
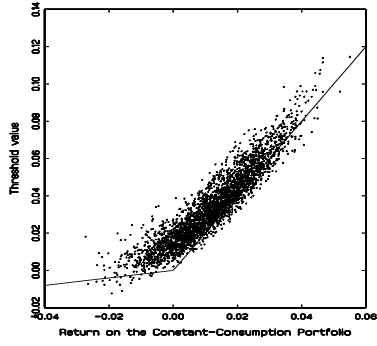
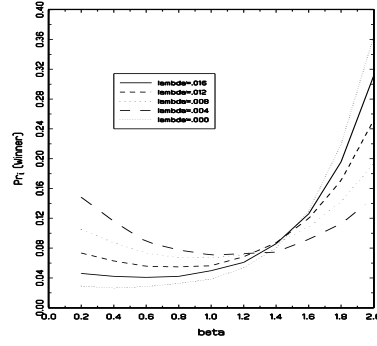


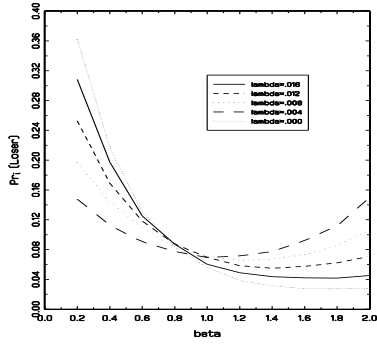
Figure 2 displays the Conditional Hansen and Jagannathan (1991) bounds on the mean and standard deviation of admissible pricing kernels for the 20 industry-sorted portfolios and the momentum portfolios. Momentum portfolios are grouped by their holding periods $K = 3, 6, 9, 12$ months.



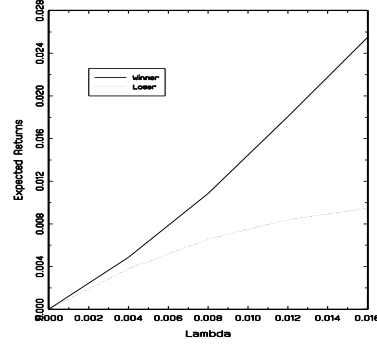
(a) The realized threshold values (based on simulation) and theoretical approximated values: $\bar{\lambda} = .012$, $\sigma_{\lambda} = .012$, and $\sigma_{\epsilon} = .010$.



(b) The functional relationship between β_{di} and Pr_i^h for different values of $\bar{\lambda}$: $\sigma_{\lambda} = .012$, and $\sigma_{\epsilon} = .010$.



(c) The functional relationship between β_{di} and Pr_i^l for different values of $\bar{\lambda}$: $\sigma_{\lambda} = .012$, and $\sigma_{\epsilon} = .010$.



(d) The relationship between the expected return on the winner (and loser) and $\bar{\lambda}$: $\sigma_{\lambda} = .012$, and $\sigma_{\epsilon} = .010$.

Figure 3: Winners, Losers, betas, and risk premium

Figure 4: Relationship Between Betas and Probabilities

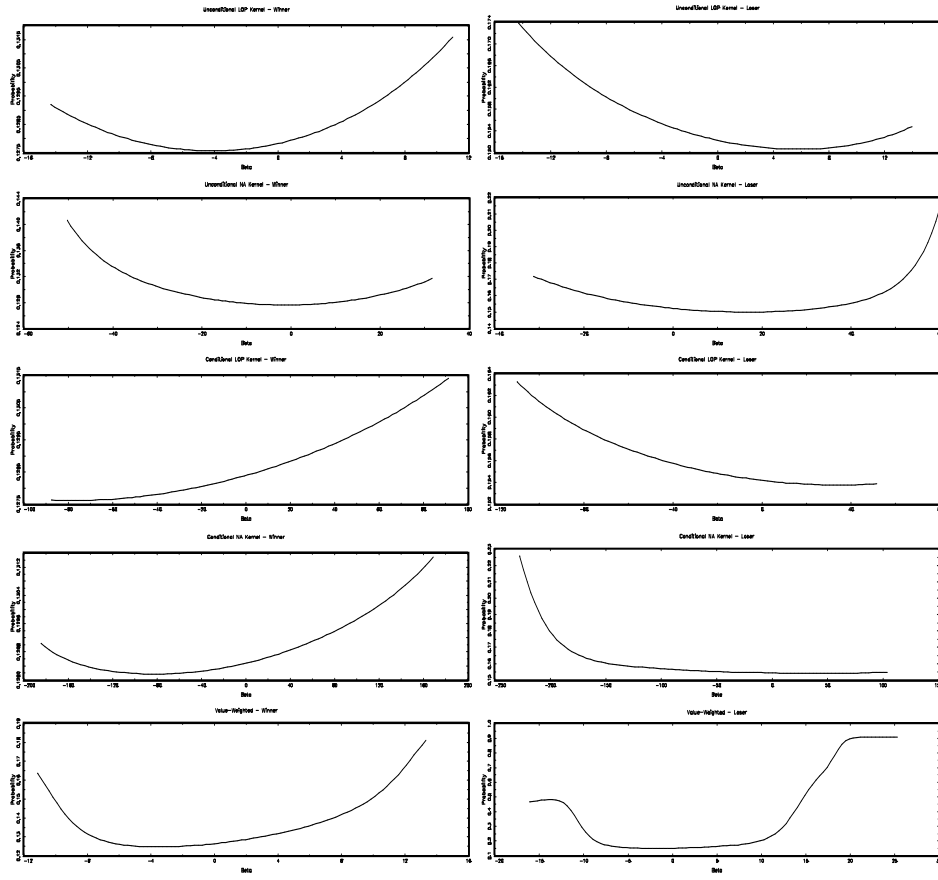


Figure 4 illustrates the nonparametric kernel estimates of the relationship between betas and probabilities.

Figure 5: Efficient Frontiers

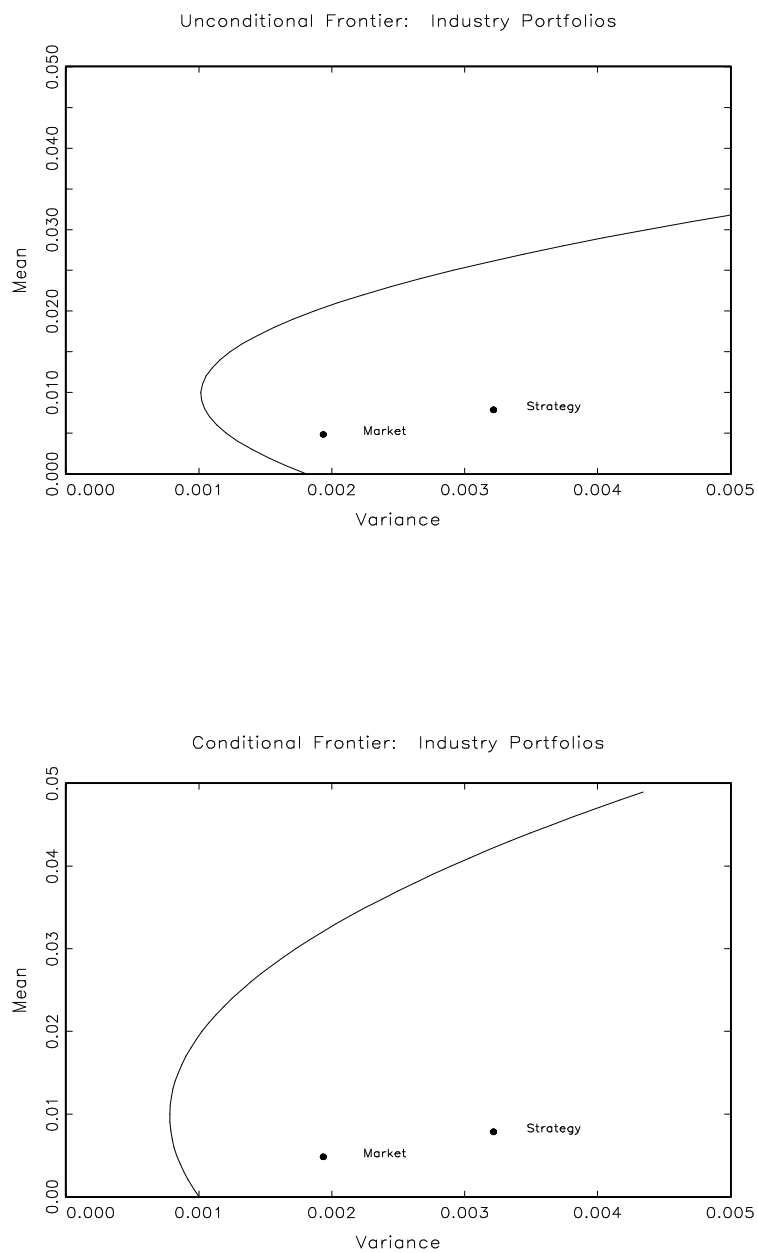


Figure 5 illustrates the unconditional and conditional frontiers constructed by the industry portfolios.